11-1 DLC SERVICES

- The data link control (DLC) deals with procedures for communication between two adjacent nodes no matter whether the link is dedicated or broadcast. Data link control functions include framing and flow and error control.

Framing

- The data-link layer needs to pack bits into frames, so that each frame is distinguishable from another.
- Our postal system practices a type of framing.
- The simple act of inserting a letter into an envelope separates one piece of information from another; the envelope serves as the delimiter.
- Framing in the data-link layer separates a message from one source to a destination by adding a sender address and a destination address.
- The destination address defines where the packet is to go; the sender address helps the recipient acknowledge the receipt.

Framing

- Whole message can be placed in a single frame, but that is not done because frame can be very large making Flow Control and Error Control very inefficient.

- When message is carried in one large frame, even single bit error would require the retransmission of the whole frame.

- When message is divided into small frames, single bit error affect only that small frame.

FrameSize

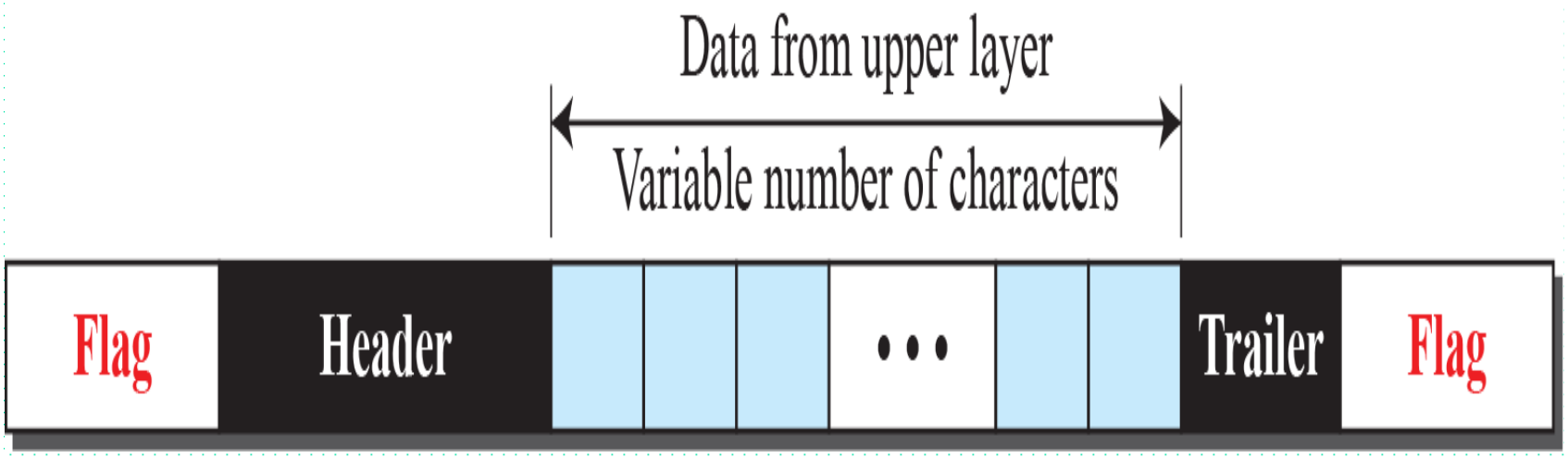
- Frame can be i) fixed size
  ii) Variable size.
- In fixed size framing, no need to define the frame boundaries. Size itself can be treated as a delimiter.
  Eg: ATM frames of fixed size called cells.

- In variable size framing, we need to define the frame boundaries to identify beginning and end of the frame.
- Two approaches are used for this purpose
  i) Character-oriented Framing
  ii) Bit-oriented Framing.

Character-oriented Framing

- Data to be carried are 8-bit characters from coding system such as ASCII.
- Header carries source, destination address along with some control information.
- Trailer carries error detection or correction redundant bits, are multiples of 8 bits.
- To separate one frame from other, a 1byte flag is added to the beginning and end of frame.
- The flag, composed of protocol dependent special characters which signals start or end of the frame.

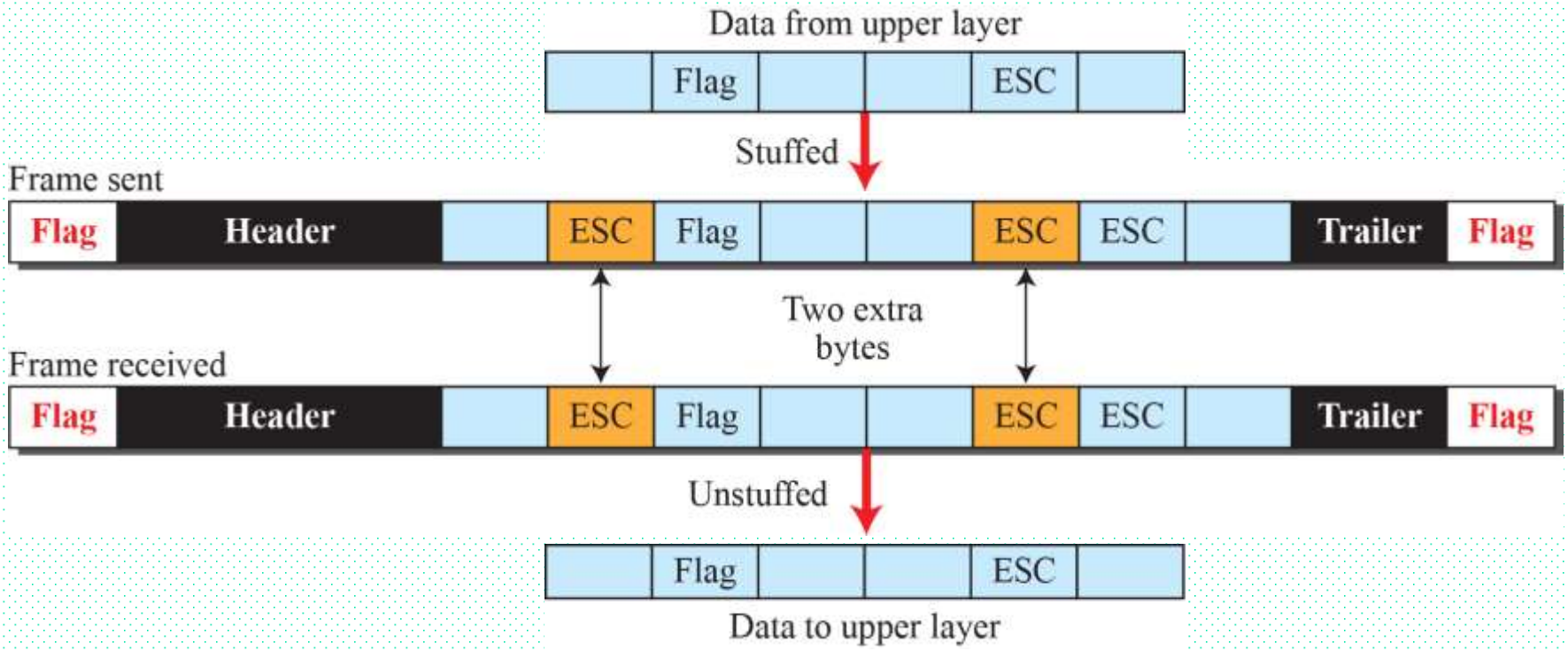
A frame in a character-oriented protocol



Character-oriented Framing

- Was popular only when text was exchanged by the DLL.
- Flag used, could be selected as any character which is not used for text communication. As we send now other types of information such graphs, audio and video, the pattern used for flag could be the part of information.
- If receiver encounters this pattern in the middle of the data, thinks it has reached the end of frame.
- Solution is byte-stuffing strategy was added in this approach.

Byte stuffing and unstuffing



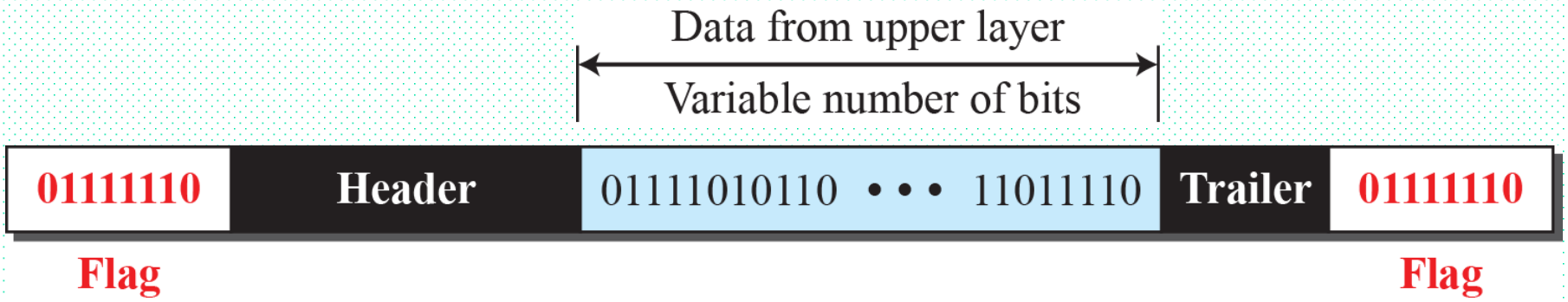
Character-oriented Framing

- In byte stuffing, a special byte is added to the data section of the frame when there is a character with the same pattern as that of the flag.
- i.e Data section is stuffed with a special extra byte called escape character. This approach presents another problem in data communication.
- The universal coding systems, unicode, in use today, have 16 or 32 bit characters that conflict with 8-bit characters.
- Therefore, next approach is used

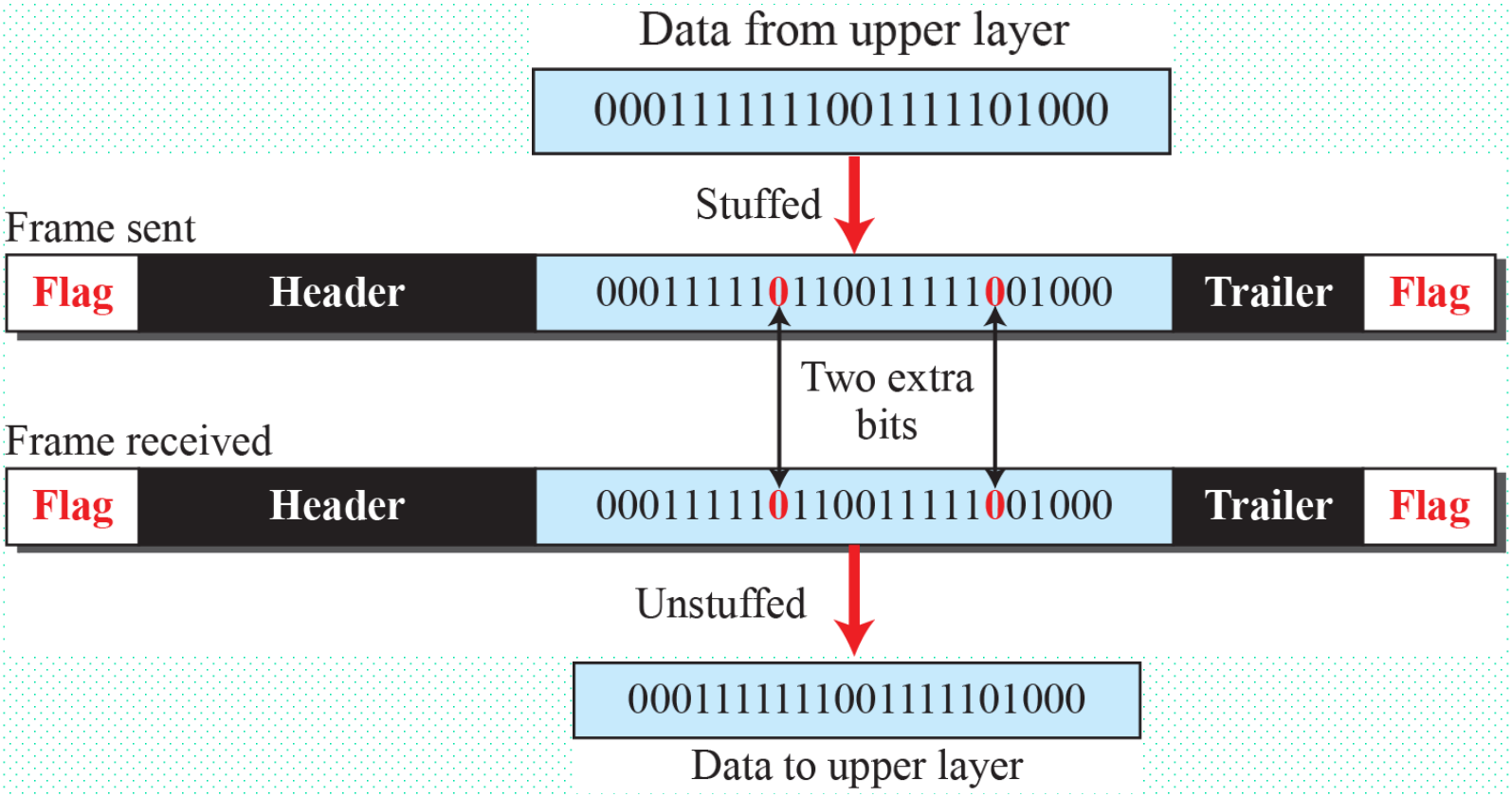
Bit-oriented Framing

- Data section of a frame is a sequence of bits to be interpreted by the upper layer as text, graphs, video and so on.
- Delimiter used to separate one frame from another and used in the beginning and end of frame is 8-bit pattern flag(01111110).
- Flag can create same problem. If flag pattern in present in the frame, need to inform the receiver that , this is data portion of the frame, not end of frame.
- This strategy is called bit stuffing.
- In bit stuffing, if a 0 and five consecutive 1 bits are encountered, an extra 0 bit added .
- Which is removed from the data by the receiver.

A frame in a bit-oriented protocol



Bit stuffing and unstuffing



Flow and Error Control

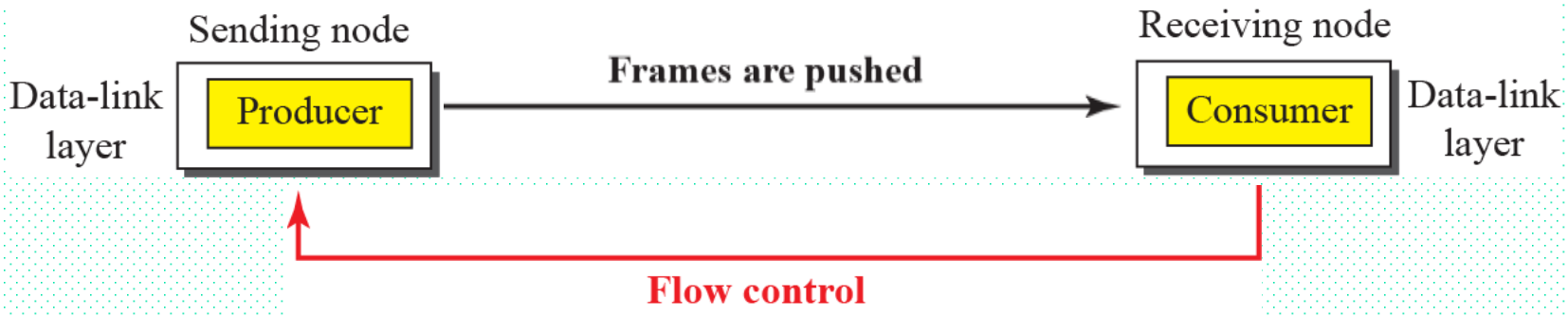
- Data Communication requires at least two devices working together, one to send and other to receive.

- For communication and exchange of information, a good coordination is must.

- Most important responsibilities of DLL are Flow Control and Error Control.

- These functions together are called data link control.

Flow control at the data link layer



Flow control

- When an entity produces items and another entity consumes them, there should be balance between them.
- If items produced faster or slower compared the rate at which it is consumed in faster/slower rate, the systems becomes less efficient.
- Flow Control is to prevent the data items at the consumer sites.
- In Data Communication (DC) at Data Link Layer (DLL), we need 4 entities:
- Network layer & DLL at sending node, and Network layer & DLL at receiving node.DLL at sending pushes frames toward DLL at receiving.”

Flow control

- If the receiving node cannot deliver the frames in the same rate to the N/W Layer, it becomes overwhelmed with the frames.
- FC mechanism should send the feedback to the sending device to stop or slow down the transmission of frames.
- At sending and receiving nodes, buffers are used at DLL.
- A buffer is a set of memory locations that can hold packets at the sender and receiver.
- FC communication occur by sending signals from consumer to producer.
- When buffer is full, it informs sender at DLL to stop pushing frames.

Error control (EC)

- Since Underlying physical N/w is not fully reliable, EC is required at the DLL to prevent receiving node from delivering corrupted packets to its N/W Layer.
- EC at DLL is implemented using one of the following methods

i) If frame is corrupted, it is silently discarded; if it is not corrupted, packet is delivered N/W layer.

ii) If frame is corrupted, it is silently discarded; if it is not corrupted, an acknowledgment is sent to the sender.

Combination of Error control & Flow control

i) Error Control & Flow Control can be combined.

ii) An Acknowledgment that is sent for flow control can also be used for error control to send the sender the packet has arrived uncorrupted.

iii) Lack of Acknowledgment means that there is a problem in the sent frame.

Connectionless and Connection-Oriented

- A DLC protocol can be either connectionless or connection-oriented.

Connectionless Protocol:

- Frames are sent from one node to another without any relationship between the frames, each frame is independent.
- Connectionless means there is no connection between the frames.
- Frames are not numbered and there is no sense of ordering.

Eg: Most of DLL protocols for LANs are connectionless protocols.

Connection-oriented Protocol

- A logical connection should be established between the two nodes (Setup phase).
- All frames that are related to each other are transmitted (Transfer phase)
- A Logical connection is terminated(teardown phase).
- Frames are numbered and sent in order.
- If they are not received in order, the receiver needs to wait until all frames belonging to the same set are received and then deliver them in order to Network Layer.

11-2 DATA-LINK LAYER PROTOCOLS

•Traditionally four protocols have been defined for the data-link layer to deal with flow and error control:

Simple, Stop-and-Wait, Go-Back-N, and Selective-Repeat.

Although the first two protocols still are used at the data-link layer, the last two have disappeared.

The behavior of DLL protocols can be better shown as a finite state machine(FSM)

Finite State Machines

- A FSM is thought of as a machine with a finite number of states.
- A machine is always in one of the state until an event occurs.
- Each event is associated with two reactions

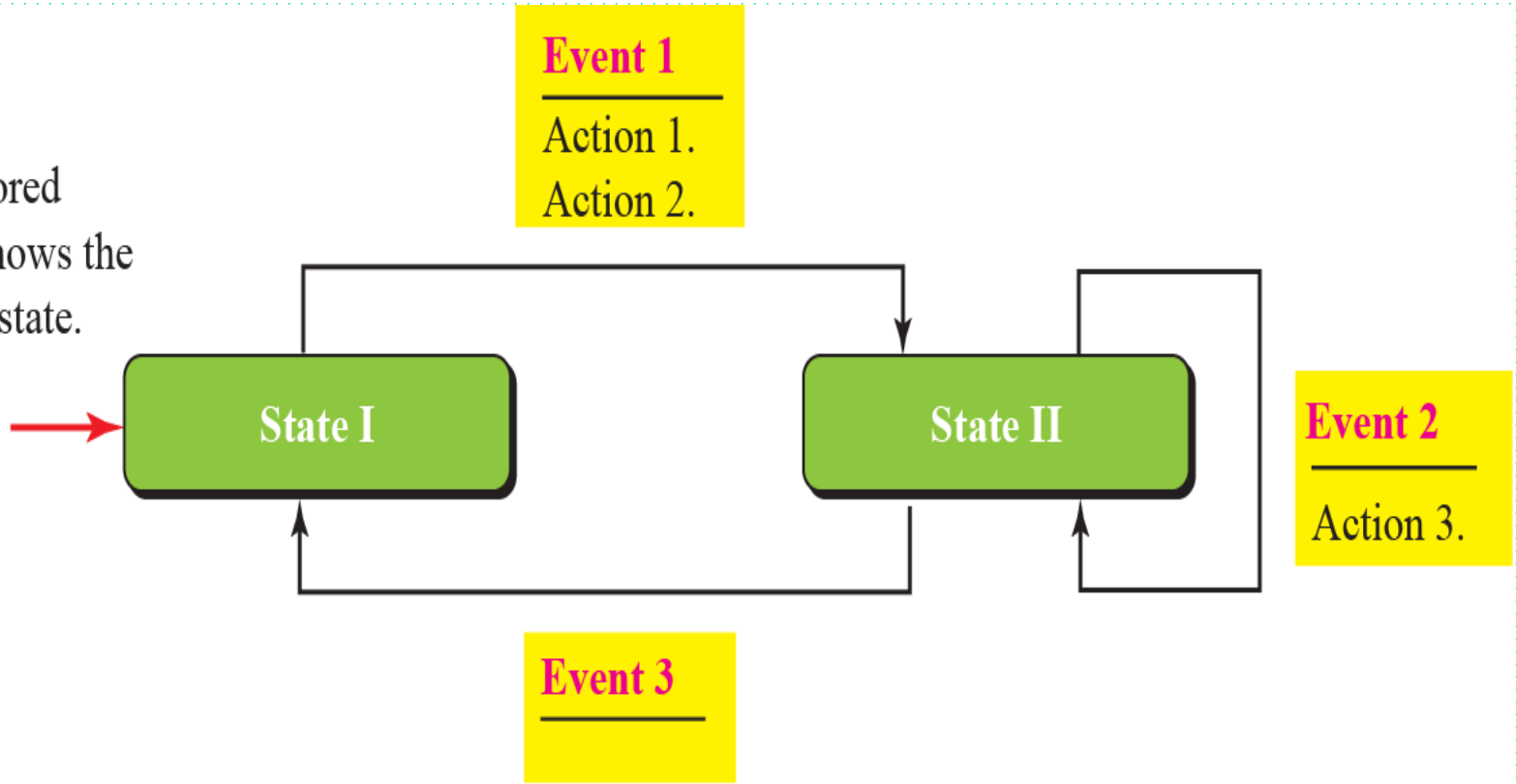
i) Defining the list of actions to be performed

ii) Determining the next state(Which can be same as current state).

- One of the state must be defined as the initial state, the state in which machine starts when turns on.

FSMs

Note:
The colored arrow shows the starting state.



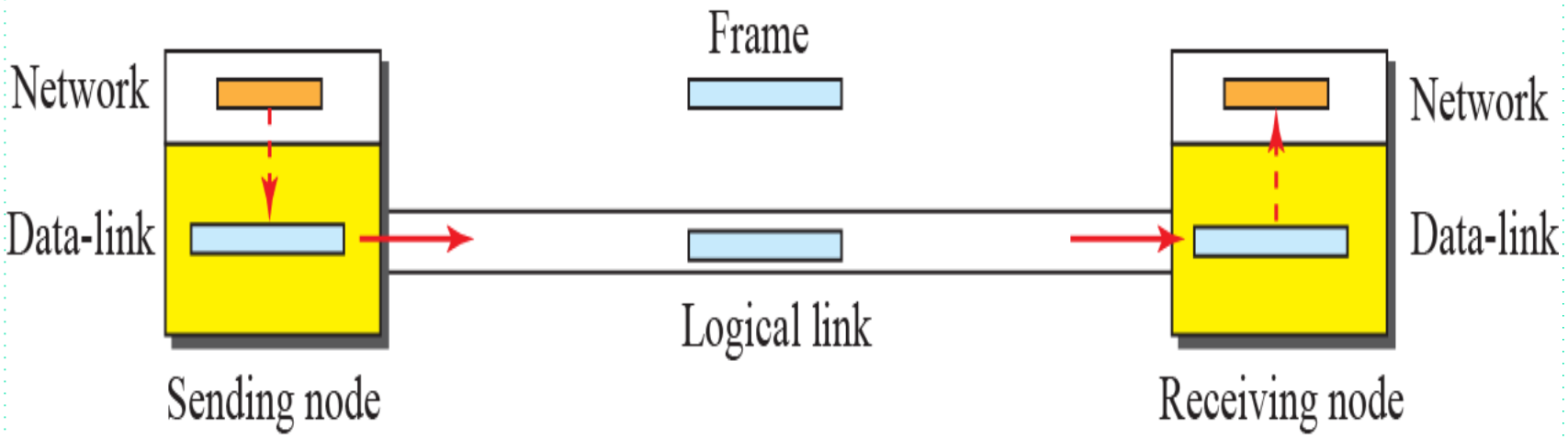
FSMs

- Machine with three states.
- There are only 3 possible events and 3 actions.
- Machine starts with state1.
- → If event1 occurs, the machine performs action 1 & 2 & enters in to state2.
- → When machine is in state2, two events may occur.
- → If event 2 occurs, the machine perform action3 and remains in the same state.
- → If event 3 occurs, machine performs no action, but moves to state 1.

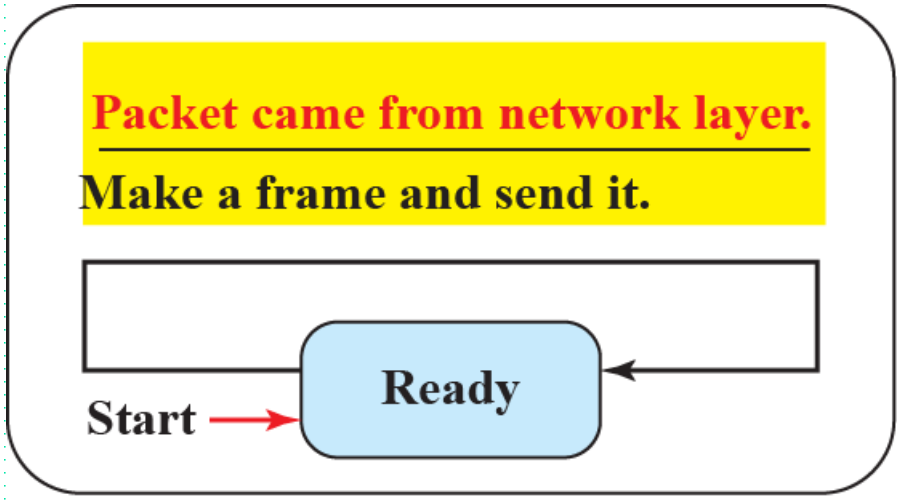
Simple Protocol

- First protocol is a simple protocol with neither flow nor error control.
- We assume that the receiver can immediately handle any frame it receives.
- In other words, the receiver can never be overwhelmed with incoming frames.
- Figure shows the layout for this protocol.
- DLL at sender gets a packet from N/W Layer, makes a frames out of it and sends the frame.
- DLL at the receiver side, extracts the packet from the frame and delivers the packet to its N/W layer.

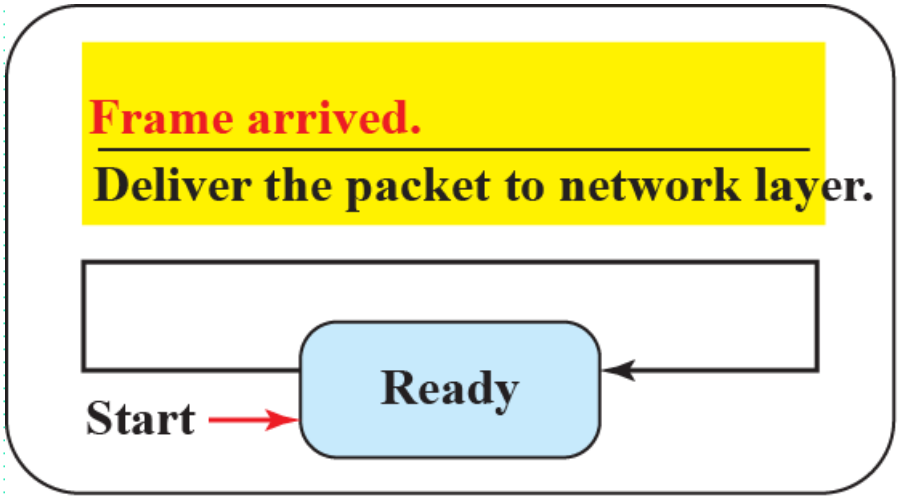
Simple protocol



FSM for the simple protocol



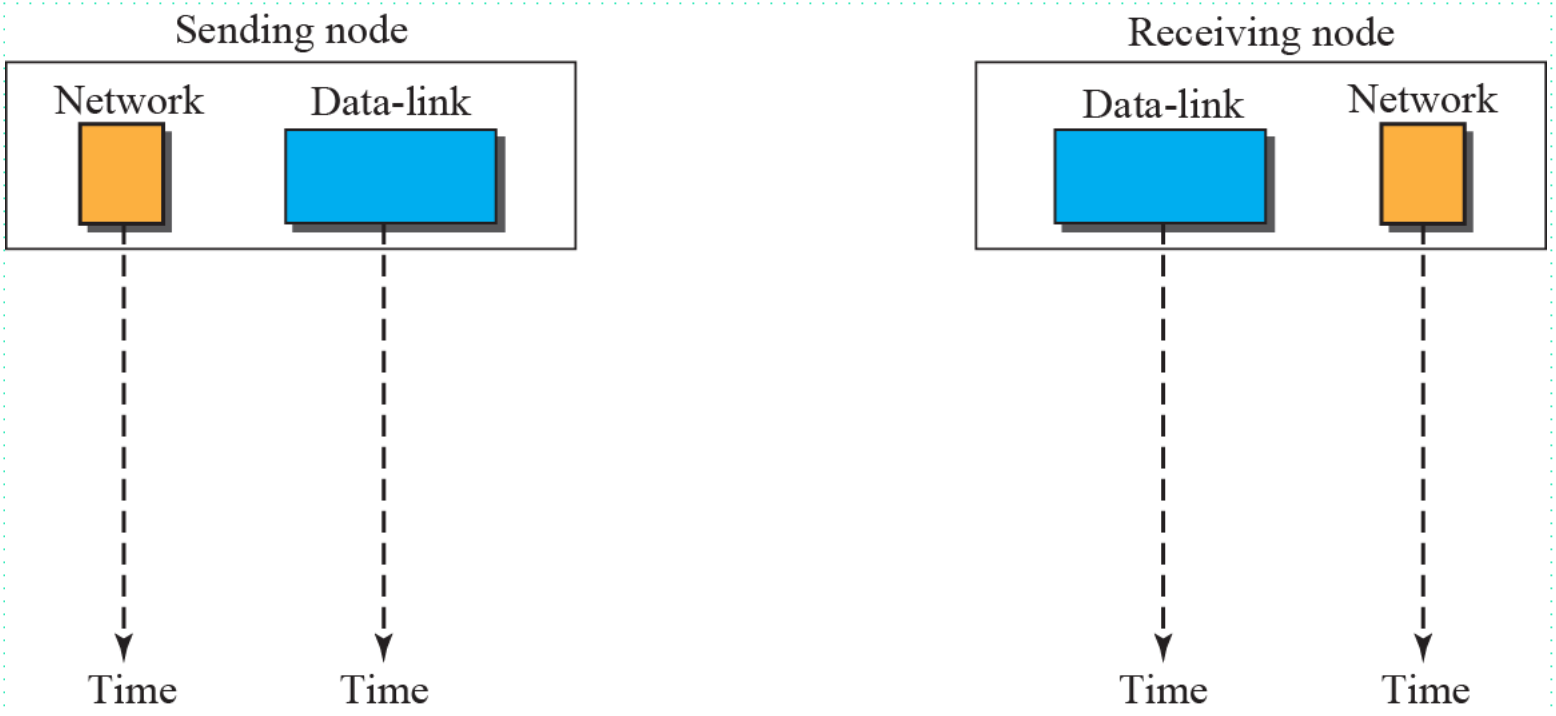
Sending node



Receiving node

Example: Figure below shows an example of communication using this protocol. It is very simple.

The sender sends frames one after another without even thinking about the receiver.



Stop-and-Wait Protocol

- Second protocol is called the Stop-and-Wait protocol, which uses both flow and error control.
- In this protocol, the sender sends one frame at a time and waits for an acknowledgment before sending the next one.
- To detect corrupted frames, we need to add a CRC to each data frame.
- When a frames arrives at the receiver site, it is checked, if its CRC is incorrect, the frame is incorrect & silently discarded.
- Silence is an alert to the sender that frame is corrupted or lost.

Stop-and-Wait Protocol (Continued)

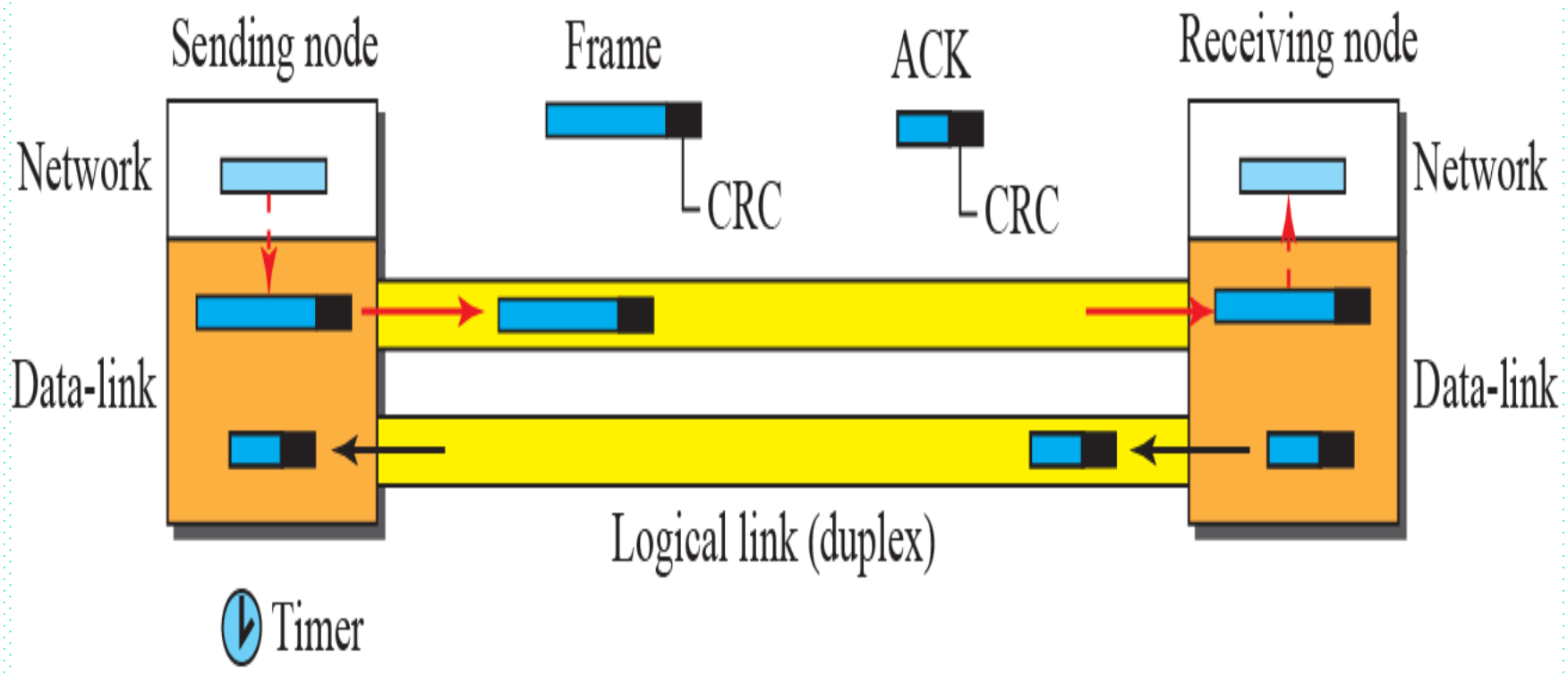
- Every time sender sends a frame, it starts a timer and if ack. arrives before timer expires, timer is stopped and next frame is transmitted.

- If timer expires before ack., frame is retransmitted because sender assumes that frame is corrupted or lost.

- This means that Sender needs to keep a copy of the sent frame until he receives the ack.

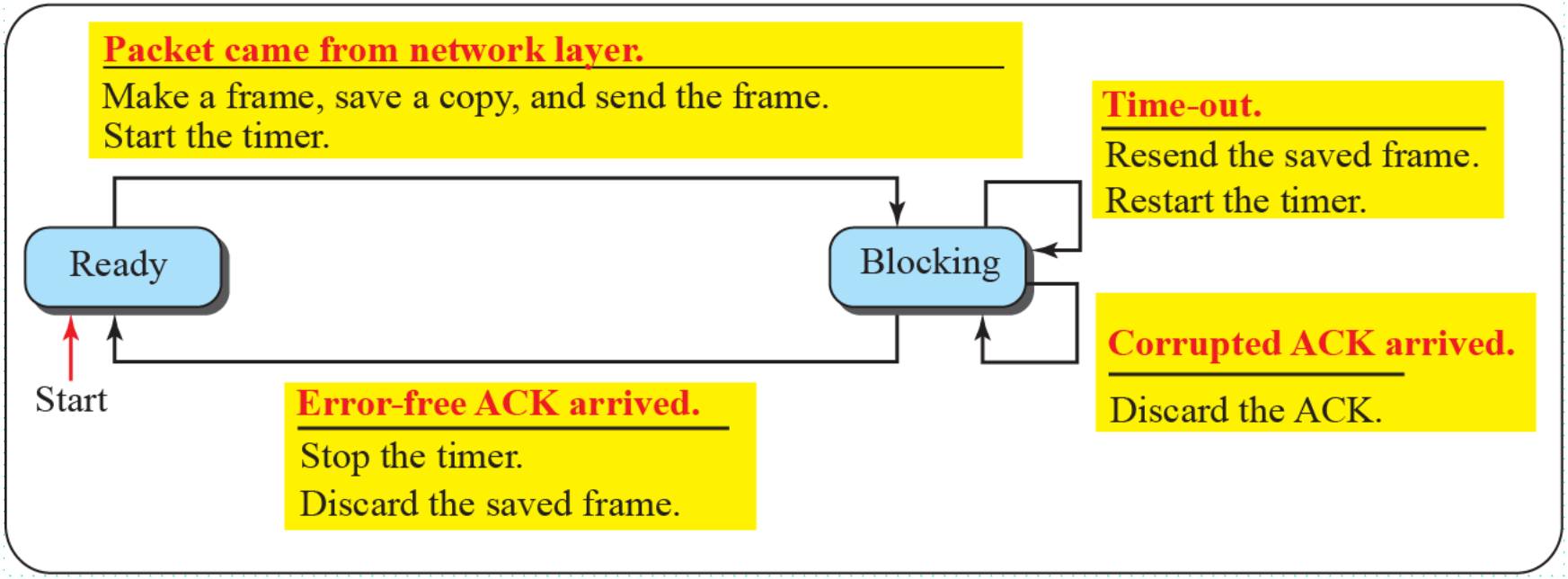
- When the corresponding acknowledgment arrives, the sender discards the frame and send the next frame if it is ready

Stop-and-wait Protocol

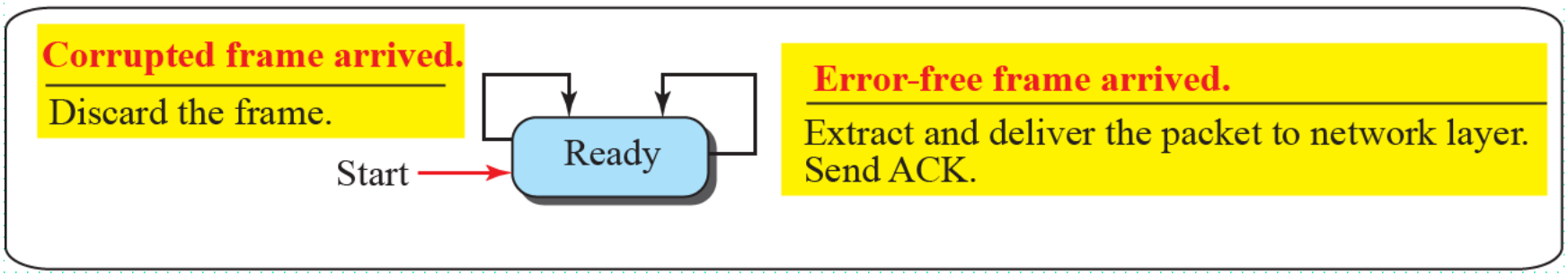


FSM for the stop-and-wait protocol

Sending node



Receiving node



FSM for the stop-and-wait protocol

- At sender side:

The sender is initially in the ready state, but it can move between the ready state and blocking state

i) Ready state: It is waiting for a packet from the N/W layer.

→Sender creates a copy of the frame, starts the timer and sends the frame. The sender moves to the blocking state.

ii)Blocking state: 3 events occur

a) if time out occurs, sender resends the saved copy of the frame and starts timer.

b) If a corrupted ACK arrives, it is discarded.

c) If an error free ACK arrives, the sender stops the timer and discards the saved copy of the frame & moves to ready state.

FSM for the stop-and-wait protocol

At receiver side:

- The receiver is always in ready state. The two events may occur

a) If an error free frame arrives, message in the frame is delivered to the network layer and ACK is sent.

b) If a corrupted frame arrives, the frame is discarded.

Example

In the figure first frame is sent and acknowledged. The second frame is sent, but lost. After time-out, it is resent.

The third frame is sent and acknowledged, but the acknowledgment is lost.

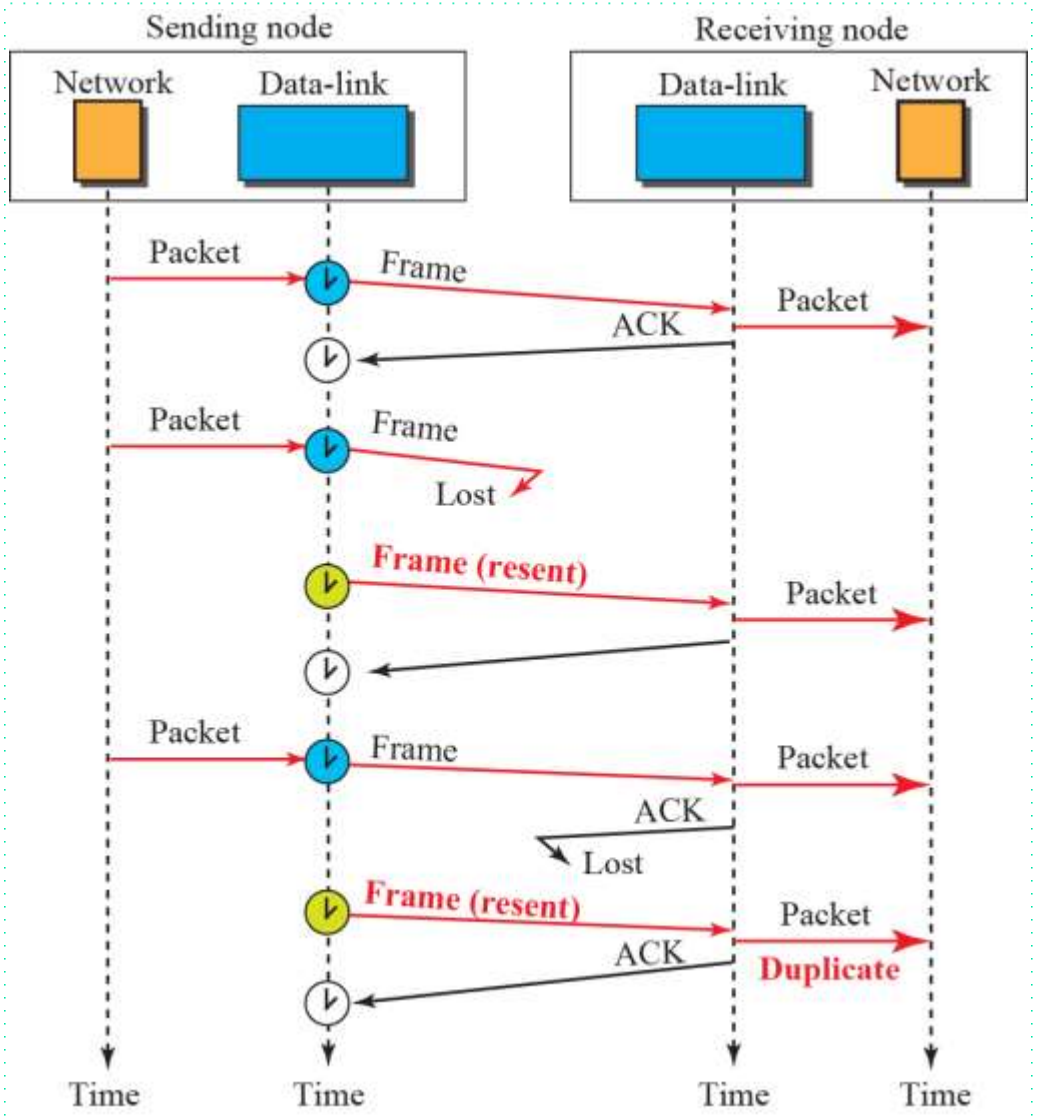
The frame is resent. However, there is a problem with this scheme.

The network layer at the receiver site receives two copies of the third packet, which is not right.

Flow diagram

Legend

- Start the timer.
- Stop the timer.
- Restart a time-out timer.



Notes:

A lost frame means either lost or corrupted.

A lost ACK means either lost or corrupted.

Figure below shows how adding sequence numbers and acknowledgment numbers can prevent duplicates.

The first frame is sent and acknowledged. The second frame is sent, but lost.

After time-out, it is resent.

The third frame is sent and acknowledged, but the acknowledgment is lost. The frame is resent.

Flow diagram

Legend

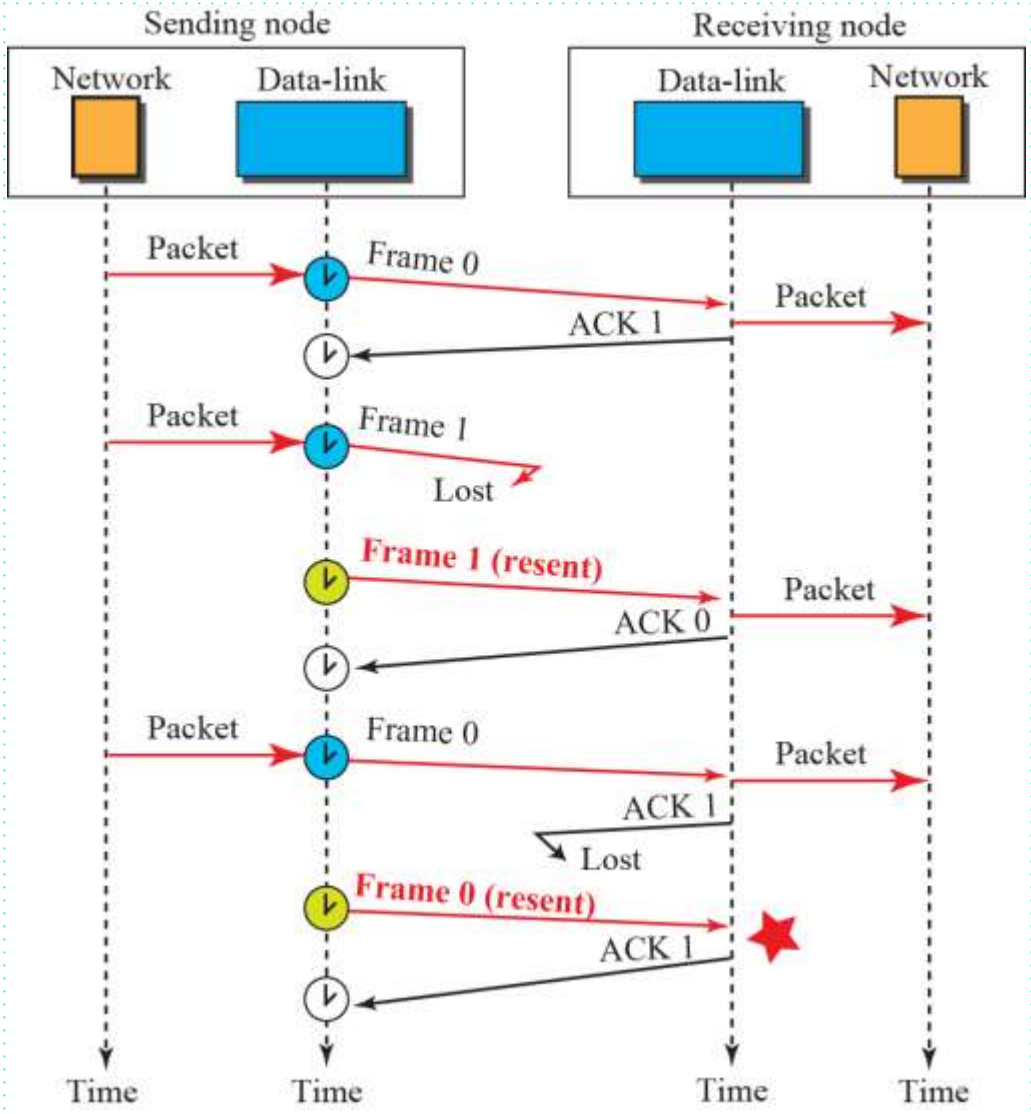
- Start the timer.
- Stop the timer.
- Restart a time-out timer.

Notes:

A lost frame means either lost or corrupted.

A lost ACK means either lost or corrupted.

Frame 0 is discarded because the receiver expects frame 1.



Piggybacking

- The two protocols we discussed in this section are designed for unidirectional communication, in which data is flowing only in one direction although the acknowledgment may travel in the other direction.

- Protocols have been designed in the past to allow data to flow in both directions.

- However, to make the communication more efficient, the data in one direction is piggybacked with the acknowledgment in the other direction.

11-4 Point to Point Protocol (PPP)

- One of the most common protocols for point-to-point access is the Point-to-Point Protocol (PPP).

- Today, millions of Internet users who need to connect their home computers to the server of an Internet service provider use PPP.

- To control and manage the transfer of data, there is a need for a point-to-point protocol at the data-link layer. PPP is by far the most common.

Services

- The designers of PPP have included several services to make it suitable for a point-to-point protocol, but have ignored some traditional services to make it simple.

Services provided by Point-point protocol

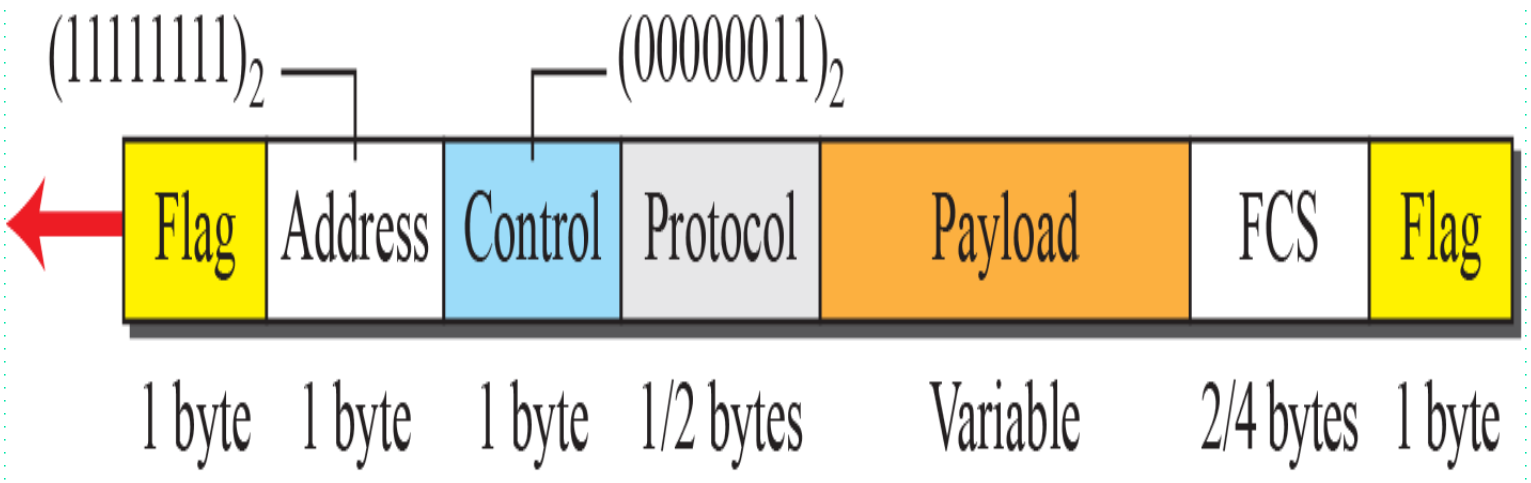
Defines

1. format of the frame to be exchanged between the devices.
2. How two devices can negotiate the establishment of the link and the exchange of data.
3. How Network layer data are encapsulated in the data link frame.
4. How two devices can authenticate each other.
5. Provides Multiple Network services supporting a variety of Network Layer protocols.
6. Connection over multiple links is provided by new version of PPP.
7. Network address configuration. Useful when a home user needs temporary Network address to connect to Internet

Services not provided by PPP

- PPP does not provide flow control. Sender can send frames one after another without concerning about overwhelming receivers buffer.
- Simple mechanism for error control. A CRC field is used to detect errors. If frame is corrupted , frame is silently discarded. Upper Layer protocol need to take care of this.
- Does not provide addressing mechanism to handle frames in a multipoint configuration.

Framing



PPP frame format

Framing

➢PPP uses a character-oriented (or byte-oriented) frame. Figure below shows the format of a PPP frame. PPP is a byte oriented protocol

Flag:→ A PPP frame starts and ends with a one byte flag with the bit pattern 0111110. The flag is treated as a byte.

Address:→ The address field is constant value and set to 11111111(broadcast address)

Control→ The field is set to a constant value 11000000. PPP doe not provide FC and limited EC. Two parties can negotiate and omit this byte.

Framing

Protocol→ defines what is being carried in the data field either user data or other information.

Default size is 2 bytes, two parties can negotiate to use only 1 byte.
Payload field→ carries use data or other info.

Data field is a sequence of bytes with the default of maximum of 1500 bytes, But can be changed during negotiation.

The data field is byte stuffed if the flag byte pattern appears in the field. Because there is no field defining the size of the data field, padding is needed if the size is less than the maximum default/negotiated value.

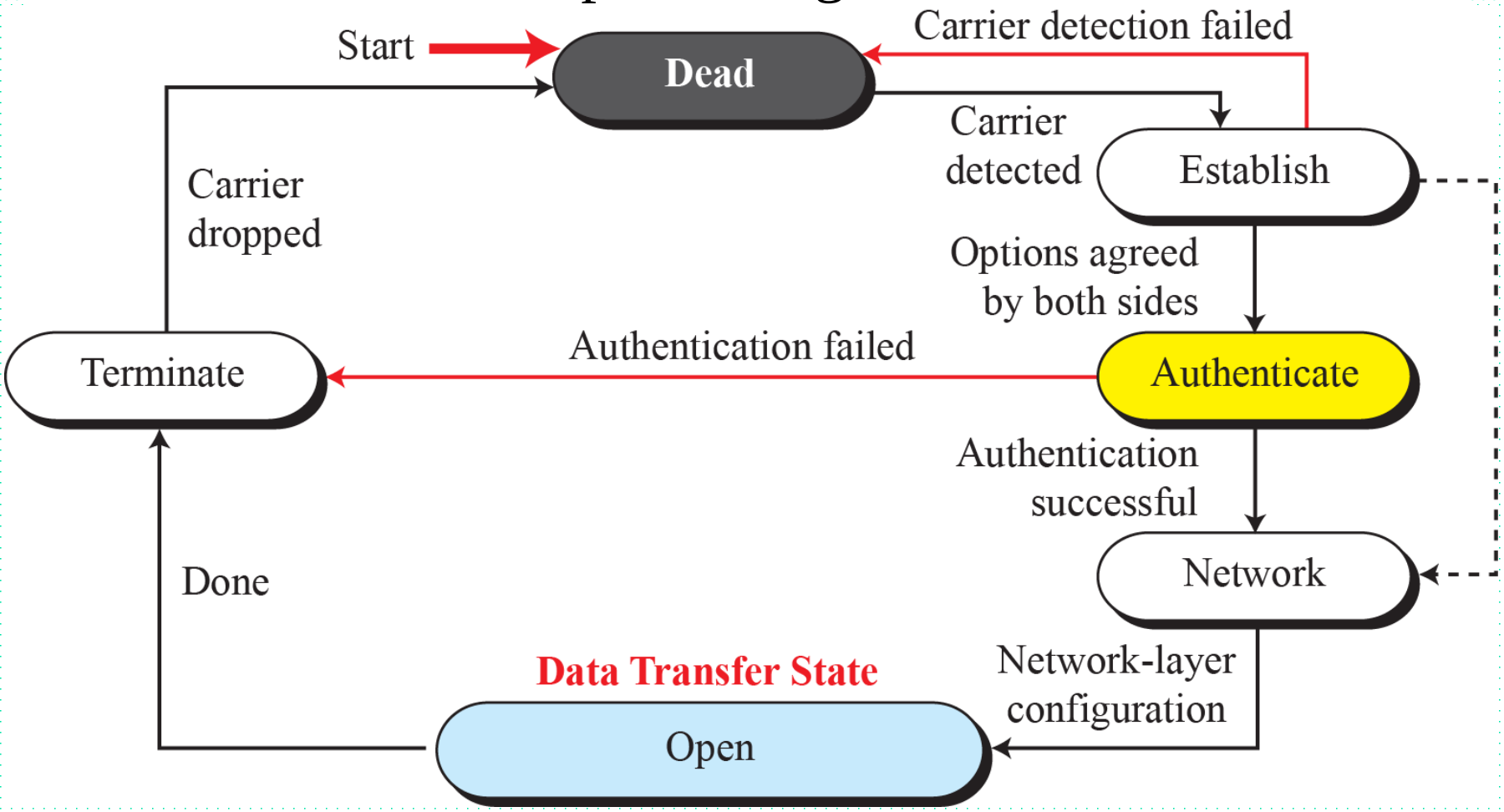
FCS:→ FCS (Frame Check Sequence) is simply a 2/4 byte standard CRC

Byte stuffing

- In PPP flag is byte. The flag byte need to be escaped whenever it appears in the data section of the frame.
- The escape byte is 01111101 , which means that every time the flag like pattern appears in the data, this extra byte is stuffed to tell the receiver the next byte is not a flag.
- PPP is a byte-oriented protocol using byte stuffing with the escape byte 01111101.

Transition Phases

➢A PPP connection goes through phases which can be shown in a transition phase diagram.



Transition phases

Dead: In this phase link is not used. There is no active carrier and line is quiet.

Establish: When one of the node starts communication, connection goes into establish phase. Options are negotiated between two parties.

If negotiation is successful, system goes to authentication phase or directly to networking phase. Link control packet are used in this phase.

Authenticate: Two nodes can negotiate and skip this phase as it is optional. If not to skip, authentication packets are used.

Transition phases

Network: Negotiation for the network layer protocols takes place. PPP specifies that two nodes establish a Network agreement before data at the Network can be exchanged.

PPP supports multi protocols at Network Layer.

If a node is running multi protocol simultaneously at the Network Layer, receiving node needs to know which protocol will receive the data.

Transition phases

Open: Data transfer takes place.

When a connection reaches this phase, exchange of data packets takes place

Connections remains in this phase until one of the endpoints wants to terminate the connection

Terminate: Connection is terminated. Several packets are exchanged between the two ends for house cleaning and closing of the link

12-1 RANDOM ACCESS

- In random-access or contention no station is superior to another station and none is assigned control over another.
- At each instance, a station that has data to send uses a procedure defined by the protocol to make a decision on whether or not to send.
- This decision depends on the state of the medium (idle or busy).
- Each station can transmit when it desires on the condition that it follows the predefined procedure, testing the state of the medium.

Random access(Contention Method)

- Two features that gave this name

1) There is no scheduled time for the station to transmit. Transmission is random among the stations.

So, the name *Random access method*

2)No rules specify which station should send next. Stations compete with one another to access the medium.

So, the method is called Contention methods

Random access

- As each station has right to access the medium, if more than one station tries to send, there is an access conflict and frames will be either modified or destroyed.
- To avoid this access conflict, each station follows the procedures which answers questions

1)When can the station access the medium?

2)What can station do if medium is busy.

3)How can the station determine the success or failure of the transmission

4) What can station do if there is access conflict

Random access

- Methods

ALOHA

CSMA/CD

CSMA/CA

ALOHA

- ALOHA, the earliest random access method, was developed at the University of Hawaii in early 1970.
- It was designed for a radio (wireless) LAN, but it can be used on any shared medium.
- It is obvious that there are potential collisions in this arrangement.
- The medium is shared between the stations. When a station sends data, another station may attempt to do so at the same time.
- The data from the two stations collide

Pure ALOHA

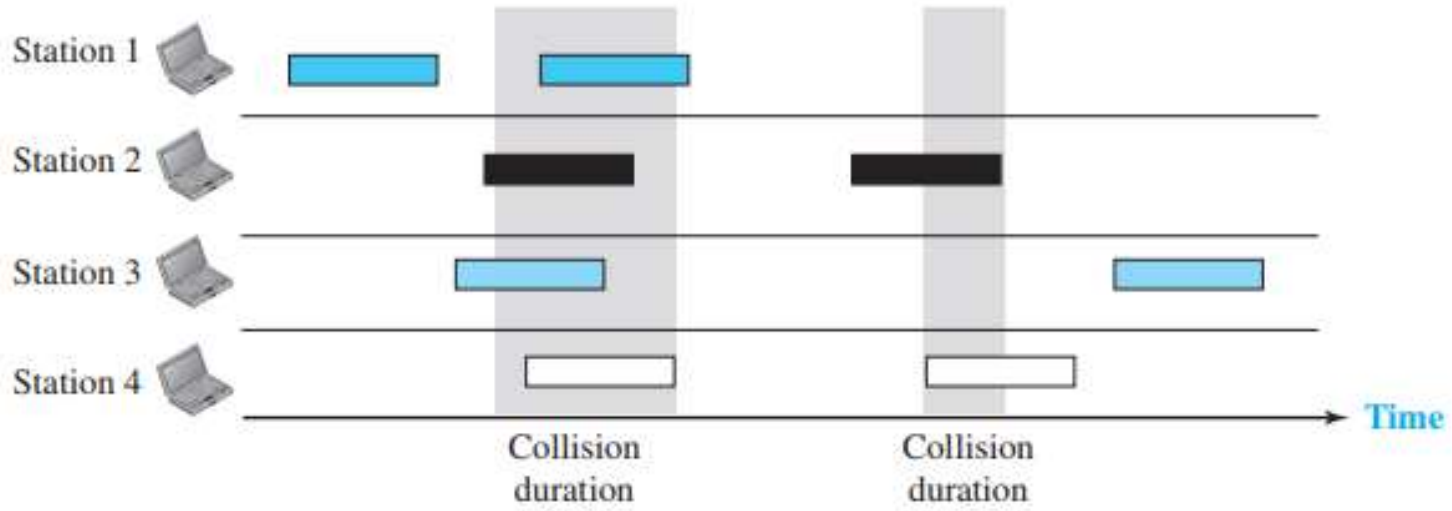
- Original ALOHA protocol is called Pure ALOHA.

Idea is

1) Each station sends a frame whenever it has frame to send.
2) Since there is only one channel to share, there is possibility of collision between frames from different stations.

Frames in a pure ALOHA network

Figure 12.2 *Frames in a pure ALOHA network*



ALOHA

- Eg: There are 4 stations that content with one another for access to the shared channel.
- Each station sends two frames, there are totally 8 frames on this medium.
- Some of these frames collide because multiple access frames are in the contention for the shared channel.
- Only two frames, frame 1.1 from station 1 and frame 3.2 from station 3 survived.

ALOHA

- Even if 1 bit of a frame coexists on the channel with one bit from another frame, there is a collision and both will be destroyed.
- Need to resend the frames that have been destroyed during transmission.
- Pure ALOHA protocol relies on Ack. from the receiver.
- If station does not receive ack. with in time-out period, station assumes that frames have been destroyed and resends it.

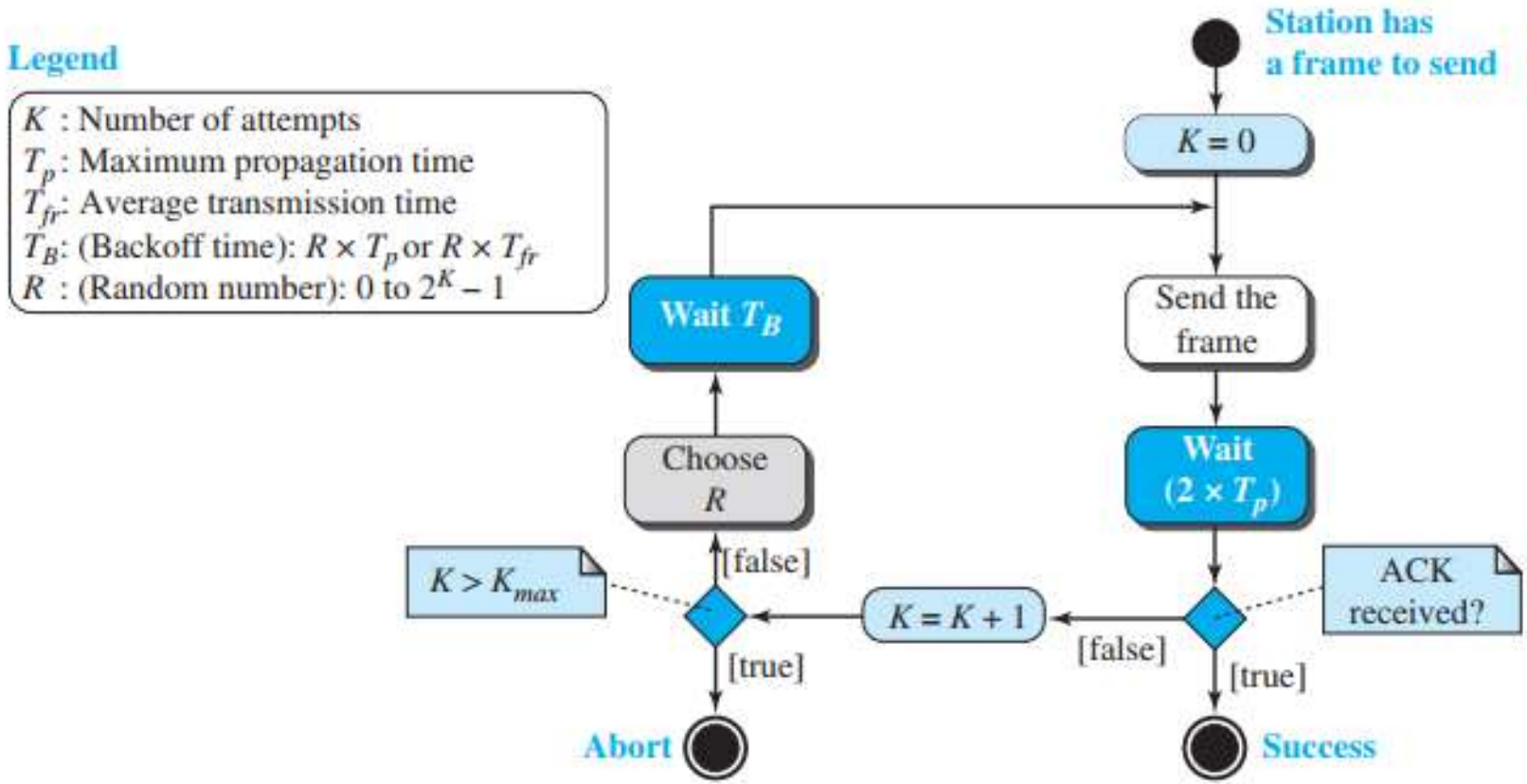
Pure ALOHA

- A Collision involves two or more stations. If all these stations resend frames again, frames collide.
- PURE ALOHA dictates that when the time out period passes, each station waits for random amount of time before resending the frames.
- Randomness avoid more collisions .
- This is called Back-off time T_B

Pure ALOHA

- Second method to prevent congesting the channel with retransmitted frames is after maximum number of retransmission attempts K_{max} the stations must give up and try later.
- Timeout period is equal to maximum possible RT (Round Trip) propagation delay, which is twice the the amount of time required to send the frame between most widely separated stations(2*Tp).
- Back-off time T_B is a random value which depends on value of K.

Figure 12.3 *Procedure for pure ALOHA protocol*



Example:

The stations on a wireless ALOHA network are a maximum of 600 km apart. If we assume that signals propagate at 3 × 10⁸ m/s, find Tp and TB.

Solution:

Propagation delay

$$T_p = \frac{distance}{propagation\ speed}$$

Distance = 600 km = $600 \times 10^3 m$*, Speed =* $3 \times 10^8 m/s$

$$T_p = \frac{600 \times 10^3}{3 \times 10^8} = \frac{600}{300000} = \frac{6}{3000} = 0.002\ s = 2\ ms$$

Backoff time

For K = 2:

$$2^2 - 1 = 3$$

So the random backoff variable:

$$R = \{0, 1, 2, 3\}$$

$$T_B = R \times T_p$$

Possible values:

•*If R = 0 → TB = 0 ms ,R = 1 → TB = 2 ms, R = 2 → TB = 4 ms, fR = 3 → TB = 6 ms So, TB can be 0, 2, 4, or 6 ms depending on R.*

Continued…

- Common formula is binary exponential back-off.

In this method, for each retransmission, a multiplier in the range 0 to 2^K-1 is randomly chosen and multiplied by T_p (Max. propagation time) or T_{fr}(Average time required to send out a frame) to find T_B

The range of random numbers increases after each collision.

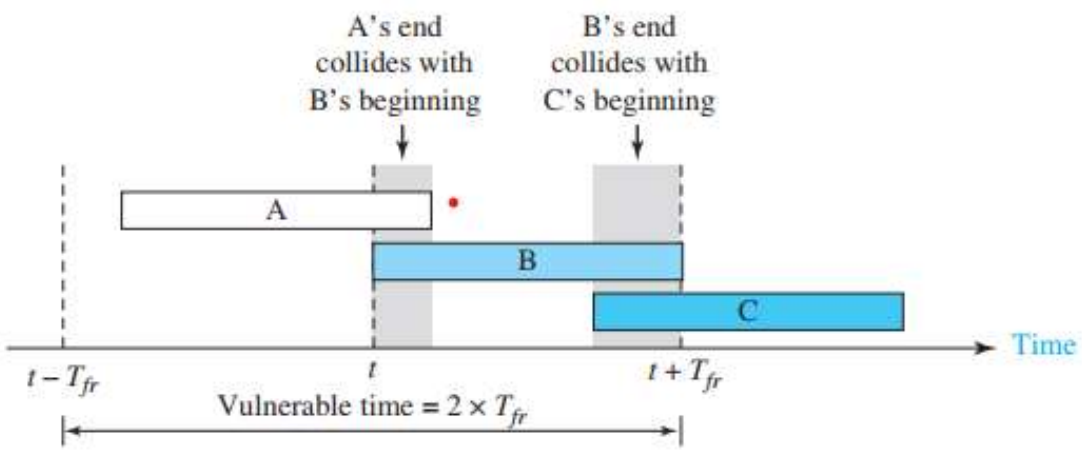
Usually the value of Kmax chosen is 15

Vulnerable time

- Vulnerable time is length of a time, in which there is a possibility of collision.
- Assume that stations send fixed frames with each frame taking T_{fr} (average frame transmission time) seconds to send .
- From the fig. the vulnerable time ,during which a collision may occur in PURE ALOHA, is 2 times the frame transmission time.

Pure Aloha Vulnerable time=2* T_{fr}

Figure 12.4 *Vulnerable time for pure ALOHA protocol*



Example : A pure ALOHA network transmits 200-bit frames on a shared channel of 200 kbps. What is the requirement to make this frame collision-free?

Solution

Average frame transmission time T_{fr} is 200 bits/200 kbps or 1 ms.

The vulnerable time is 2×1 ms = 2 ms.

This means no station should send later than 1 ms before this station starts transmission and no station should start sending during the period (1 ms) that this station is sending.

Throughput

- Let G be the average number of frames generated by the system during one frame transmission time.
- It can be proved that, average number of successful transmission for pure ALOHA is
- The Maximum throughput S_{max} =0.184 for G=1/2.

$$S = G * e^{-2G}$$

- *The throughput for pure ALOHA is* $S = G \times e^{-2G}$.
- *The maximum throughput*

S_{max} *= 0.184 when G= (1/2).*

Example : A pure ALOHA network transmits 200-bit frames on a shared channel of 200 kbps. What is the throughput if the system (all stations together) produces

a. 1000 frames per second?

b. 500 frames per second?

c. 250 frames per second?

Solution

The frame transmission time is 200/200 kbps or 1 ms.

a. If the system creates 1000 frames per second, then 1 frame per millisecond. Therefore $G = 1$.

Throughput $S = G \times e^{-2G} = 0.135$ (13.5 percent).

This means that the throughput is $1000 \times 0.135 = 135$ frames. Only 135 frames out of 1000 will probably survive.

b. If the system creates 500 frames per second, or 1/2 frames per millisecond.

Therefore G = 1/2.

Throughput $S=G\times e^{-2G}=0.184$ (18.4 percent).

This means that the throughput is

$500 \times 0.184 = 92$ and that only 92 frames out of 500 will probably survive.

c. If the system creates 250 frames per second, or 1/4 frames per millisecond.

Therefore G = 1/4.

Throughput $S = G \times e^{-2G} = 0.152$ (15.2 percent).

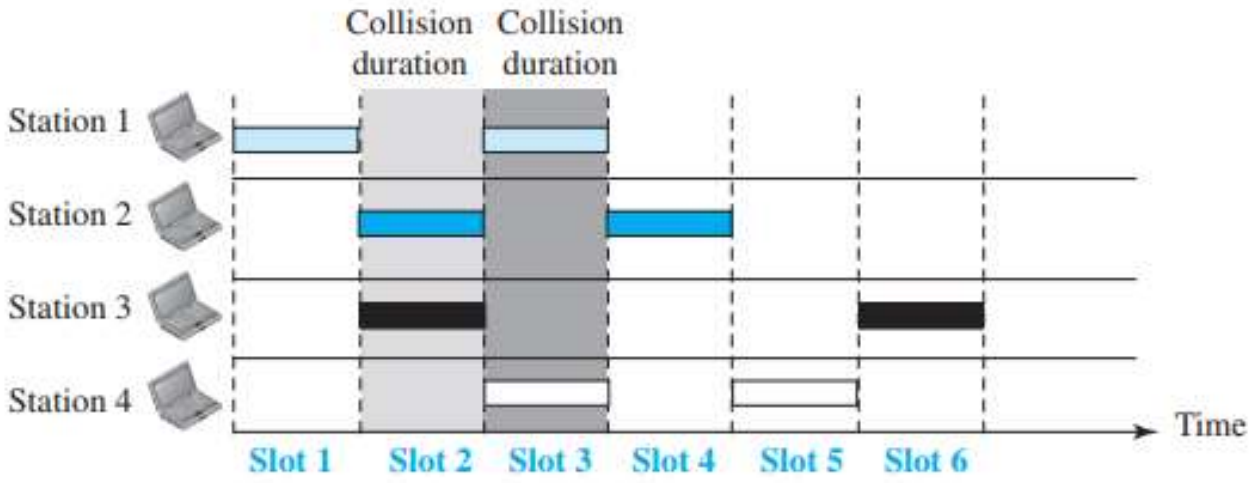
This means that the throughput is $250 \times 0.152 = 38$. Only 38 frames out of 250 will probably survive

Slotted ALOHA

- Pure Aloha vulnerable time is 2*Tfr, because no rule that defines when the station is allowed to transmit.
- Station may send before other station has finished/another station has started.
- To improve the efficiency of Pure ALOHA, Slotted ALOHA was invented.
- In Slotted ALOHA, divide the time into slots of Tfr Sec and force the station to send only in the beginning of the time slot.

Frames in a slotted ALOHA network

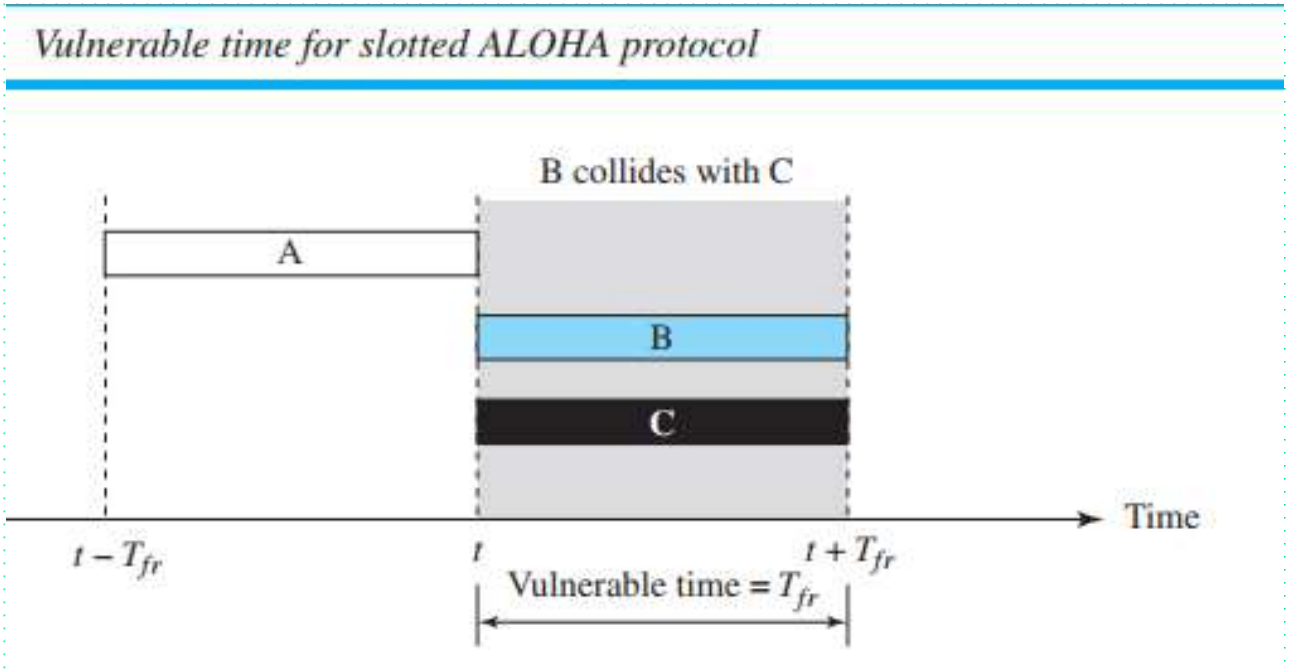
Figure 12.5 *Frames in a slotted ALOHA network*



Slotted ALOHA

- Because station is allowed to send only at the beginning of the synchronized time slot,
- if a station misses this moment, it must wait until the beginning of the next time slot.
- Still there is possibility of collision if two stations try to send at the beginning of the same time slot.
- Vulnerable time is one-half that of PURE ALOHA= T_{fr}.

Vulnerable time for slotted ALOHA protocol



Only transmissions in the same slot can collide

Transmissions in earlier or later slots cannot collide with A

Throughput

- The throughput for slotted ALOHA is

$$S = G \times e^{-G}.$$

- The maximum throughput

$$S_{max} = 0.368 \text{ when } G = 1.$$

- If one frame is generated during one frame transmission time, then 36.8% of these frames reach the destination successfully.
- Because vulnerable time is equal to frame transmission time., station generates only one frame in this time.

Example: A slotted ALOHA network transmits 200-bit frames using a shared channel with a 200-kbps bandwidth. Find the throughput if the system (all stations together) produces

a. 1000 frames per second.

b. 500 frames per second.

c. 250 frames per second.

Solution

The frame transmission time is 200/200 kbps or 1 ms.

a) G =1. So throughput $S = G \times e^{-G} = 0.368$ (36.8 percent).

This means that the throughput is $1000 \times 0.368 = 368$ frames. Only 368 out of 1000 frames will probably survive.

b) G = 1/2.

In this case $S = G \times e^{-G} = 0.303$ (30.3 percent).

This means that the throughput $500 \times 0.303 = 151.5$. Only 151 frames out of 500 will probably survive.

c) G = 1/4.

In this case $S = G \times e^{-G} = 0.195$ (19.5 percent).

This means that the throughput is $250 \times 0.195 = 49$. Only 49 frames out of 250 will probably survive.

CSMA

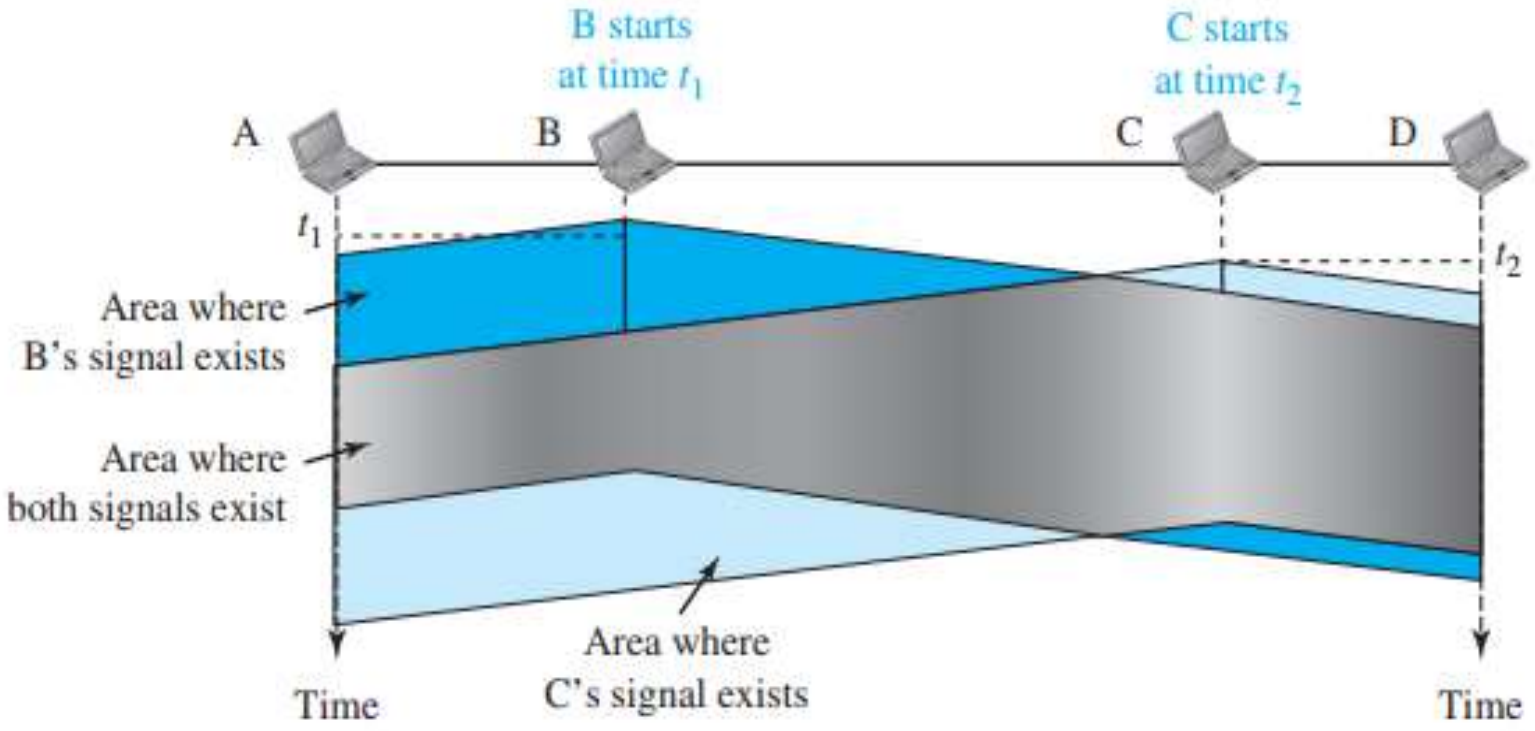
- To minimize the chance of collision and, therefore, increase the performance, the CSMA method was developed.

- The chance of collision can be reduced if a station senses the medium before trying to use it.
- Carrier sense multiple access (CSMA) requires that each station first listen to the medium (or check the state of the medium) before sending.

- In other words, CSMA is based on the principle "sense before transmit" or "listen before talk."

Space/time model of a collision in CSMA

Figure 12.7 *Space/time model of a collision in CSMA*

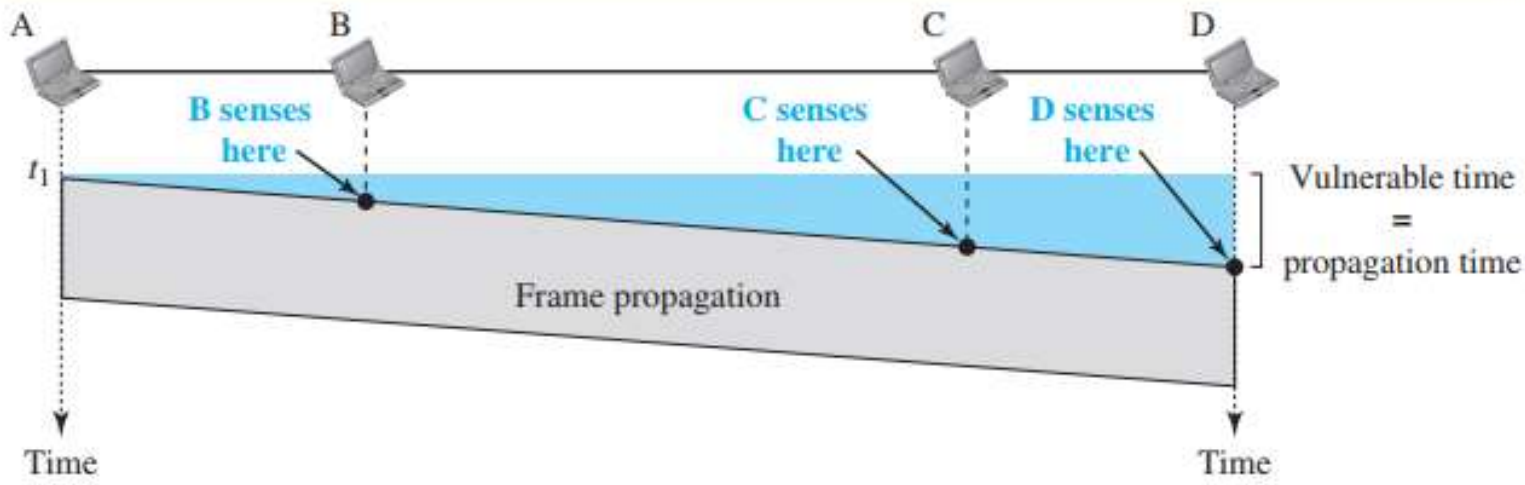


Vulnerable time

- Vulnerable time for CSMA is the Propagation time Tp.
- Tp is the time needed for the signal to propagate from one end of the medium to other.
- When any station sends a frame during this time ,collision will occur.
- If the first bit of the frame reaches the end of the medium, every station will already have heard the bit and will refrain from sending.

Vulnerable time in CSMA

Figure 12.8 *Vulnerable time in CSMA*

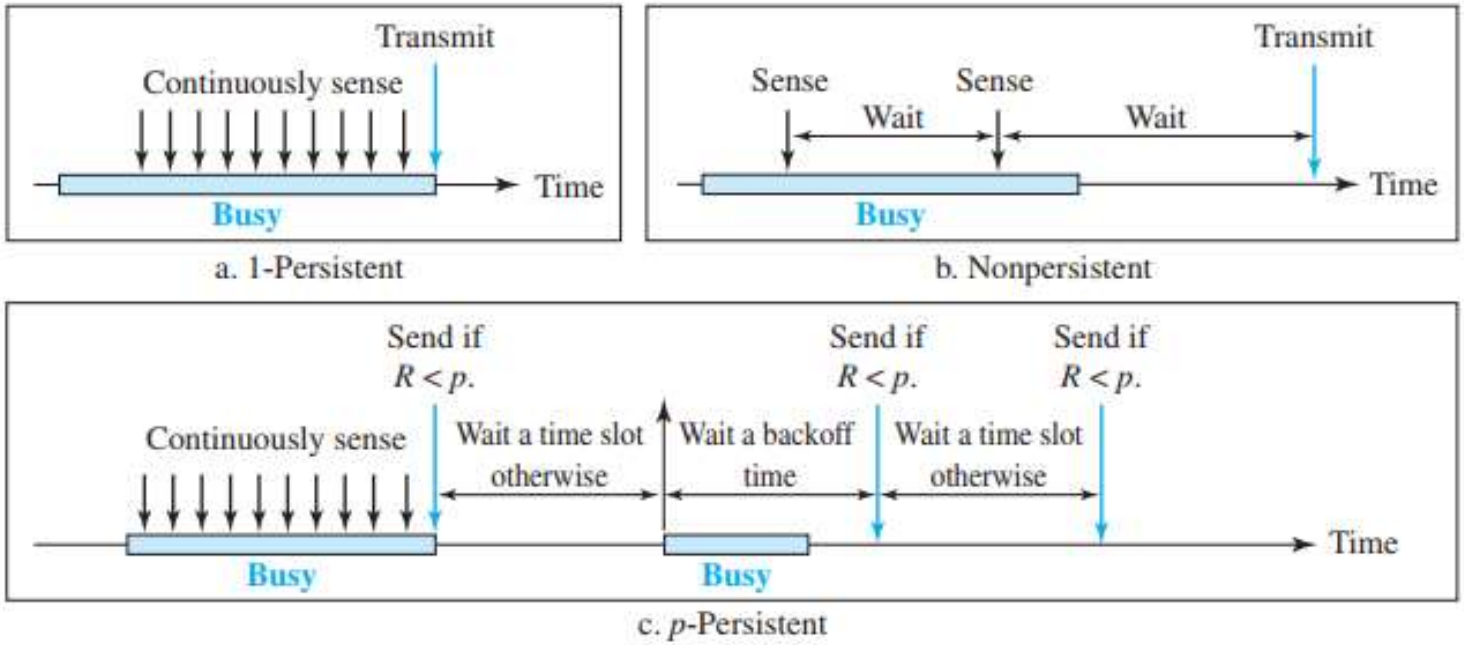


Persistence Methods

What should station do if channel is busy/idle?

- Three methods to answer this
  1) 1-Persistent method
  2) Non persistent method.
  3) p-Persistent method
- When a station finds channel is busy, behavior of three persistence methods are shown below

Figure 12.9 *Behavior of three persistence methods*



Behavior of three persistence methods

1-Persistent : After station finds the line idle, it sends the frame immediately.(Probability 1)

This methods has highest chance of collision because two or more stations may find the line idle and send their frames immediately.

Non-Persistent: Station that has frame senses the line. If line is idle, it sends immediately.

If not, it waits for random amount of time and then senses the line again.

Continued..

P-persistent: This method is used if the channel has time slot with the slot duration equal to or greater than maximum propagation time

It reduces the chance of collision and improves the efficiency.

After station finds channel is idle,

it follows the following steps

With probability p, the station sends its frame.

With probability q=1-p, the station waits for the beginning of the next time slot and checks the line again.

If the line is busy, it acts as though a collision has occurred and uses the back off procedure.

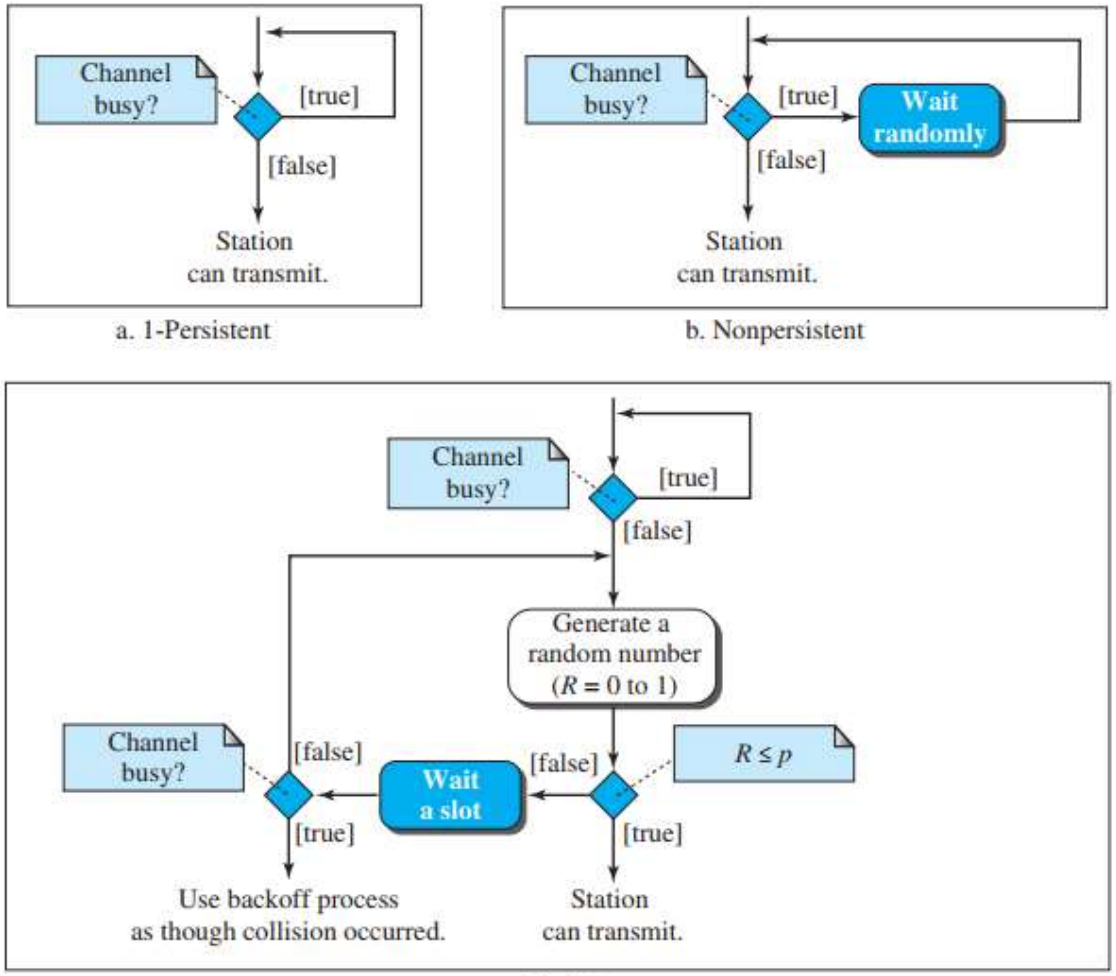
Continued..

The Non-Persistent approach reduces the chance of collision because it is unlikely that two or more stations will wait for same amount of time.

This method reduces efficiency of the n/w because the medium remains idle when there may be stations with frames to send.

Flow diagram for three persistence methods

Figure 12.10 *Flow diagram for three persistence methods*



CSMA/CD

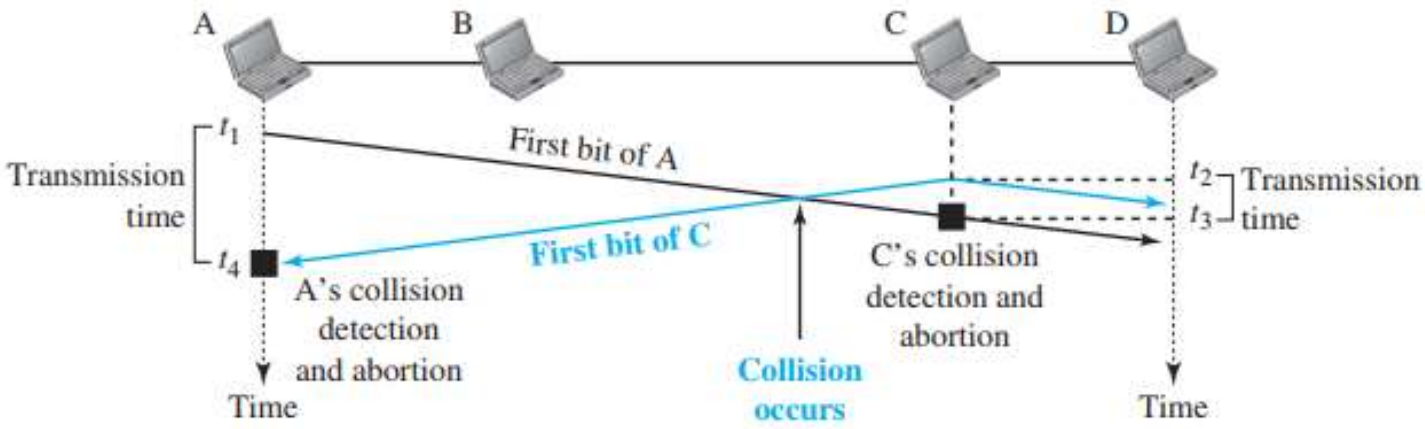
- The CSMA method does not specify the procedure following a collision.

- Carrier sense multiple access with collision detection (CSMA/CD) augments the algorithm to handle the collision.

➢ In this method, a station monitors the medium after it sends a frame to see if the transmission was successful. If so, the station is finished. If, however, there is a collision, the frame is sent again.

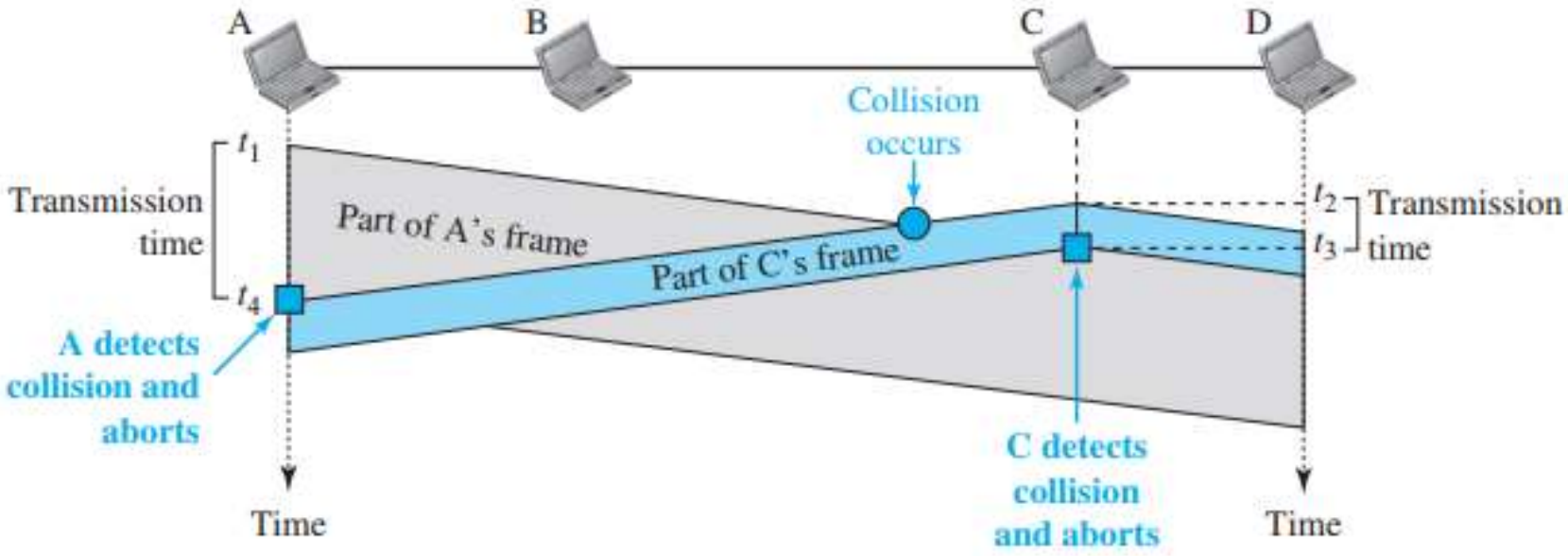
Collision of the first bits in CSMA/CD

Figure 12.11 *Collision of the first bits in CSMA/CD*



Collision and abortion in CSMA/CD

Figure 12.12 *Collision and abortion in CSMA/CD*

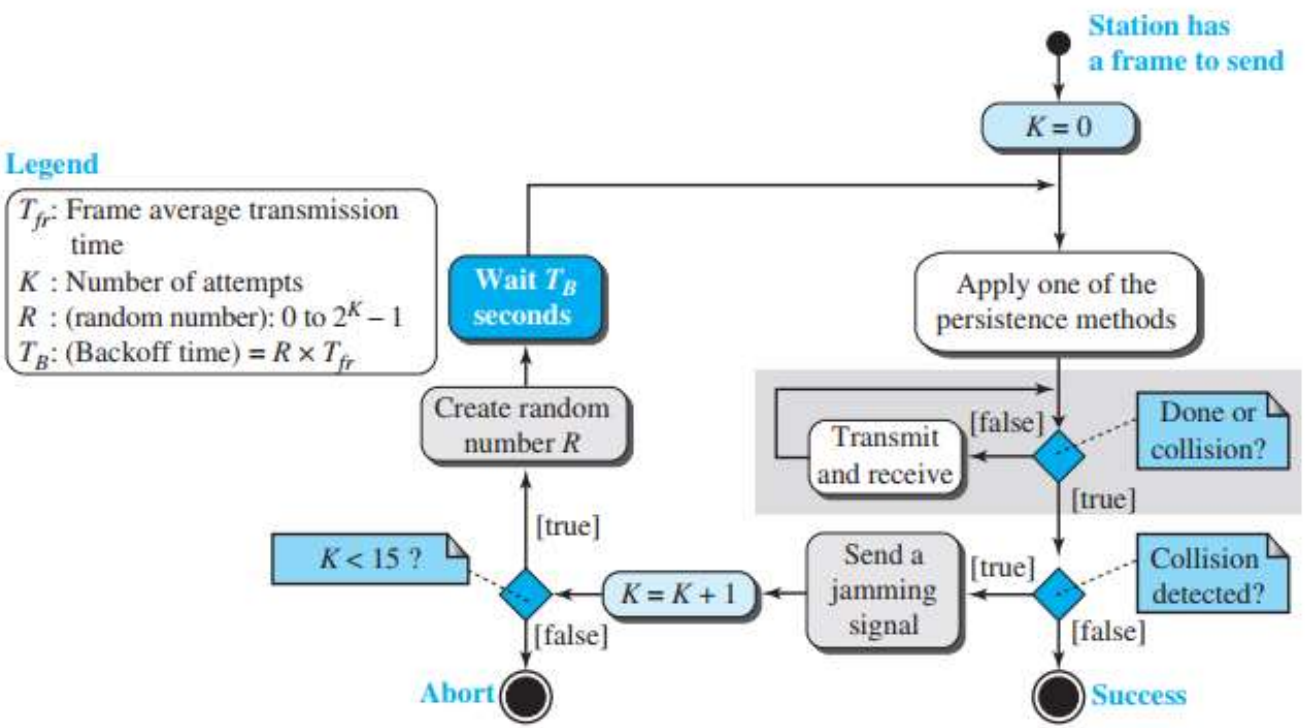


Minimum Frame size

- CSMA/CD to work, need a restriction on the frame size.
- Before sending the last bit of the frame, if sending station detects a collision, if any, abort the transmission.
- Because station once sends entire frame, does not keep a copy of the frame and monitor the line for collision detection.
- There fore frame transmission time Tfr must be at least two times the maximum propagation time Tp(2Tp).

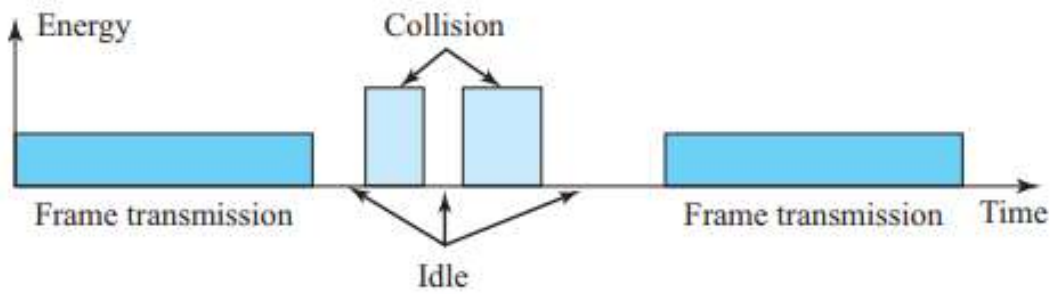
Flow diagram for the CSMA/CD

Figure 12.13 *Flow diagram for the CSMA/CD*



Energy level during transmission, idleness, or collision

Figure 12.14 *Energy level during transmission, idleness, or collision*



Energy level

- Level of energy in a channel can have three values zero, normal and abnormal.
- At Zero level, channel is idle.
- At Normal Level, station has successfully captured the channel and is sending its frame.
- At Abnormal level, there is a collision and level of energy is twice the normal level.
- The station that has frame to send, needs to monitor the energy level of the channel.

Throughput

- Is greater than PURE ALOHA.
- Maximum throughput occurs at a different values of G and is based on Persistence method and value p in p-persistent approach.
- For 1-persistent method the maximum through put is around 50% when G=1.
- For non persistent method, the maximum through put can go up to 90% when G is between 3 and 8.

CSMA/CA

- Carrier sense multiple access with collision avoidance (CSMA/CA) was invented for wireless networks.
- Collisions are avoided through the use of CSMA/CA's three strategies:

1) Inter frame space,
2) Contention window
3) Acknowledgments

Inter frame Space(IFS)

- Collisions are avoided by deferring the transmission even if channel is idle.
- Station does not send immediately, it waits for a period called IFS.
- Even though channel may be found idle when it is sensed, a distant station may have already started transmitting and the signal has not yet reached this station.
- IFS time allows signal from the distant station to reach this station.

Inter frame Space(IFS)

- After IFS time, if channel is found idle, the station can send, but still need to wait a time equal to contention time.
- IFS variable can be used to prioritize the stations.

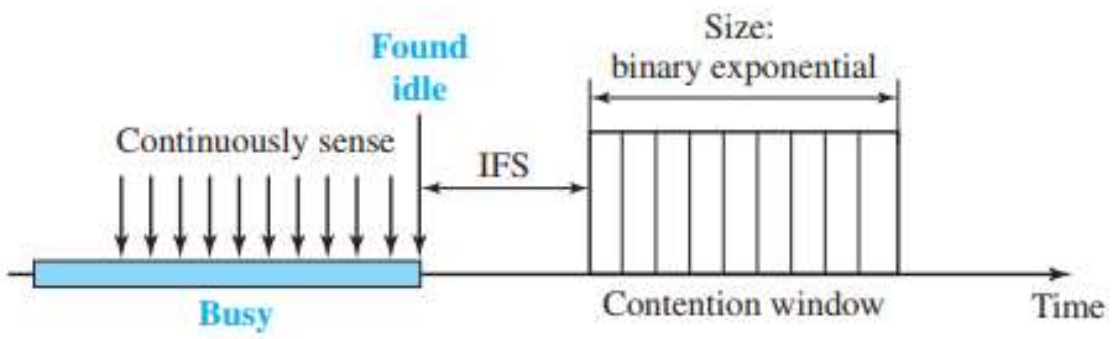
In CSMA/CA, the IFS can also be used to define the priority of a station or a frame.

Contention window

- Is an amount of time divided into slots.
- A station that is ready to send chooses a random number of slots as its wait time.
- The number of time slots in the window changes according to binary exponential back off strategy.
- Means it is set to one slot first time and then doubles each time the station can not detect an idle channel after IFS time.

Contention window

Figure 12.16 *Contention window*



Contention window

- Station needs to sense the channel after each time slot.
- If station finds that channel is busy, it does not restart the process; stops the timer and restarts it when the channel is sensed as idle.
- This gives priority to the station with longest waiting time.

In CSMA/CA, if the station finds the channel busy, it does not restart the timer of the contention window;

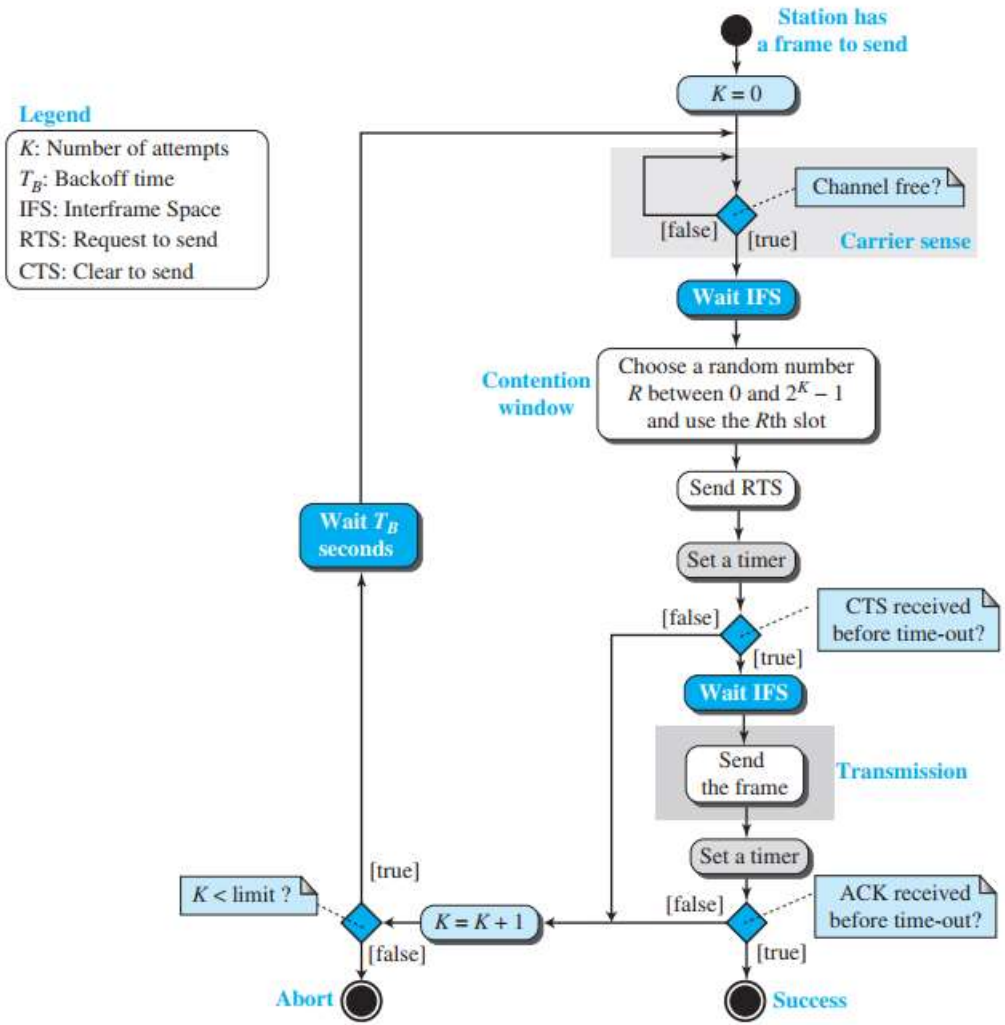
it stops the timer and restarts it when the channel becomes idle.

Acknowledgment

- With all these precautions, data may get corrupted during transmission resulting in destroyed data.
- Positive ack. and the time-out timer guarantee that receiver has received the frame.

Flow diagram for CSMA/CA

Figure 12.15 *Flow diagram of CSMA/CA*



Frame exchange Time Line

1. Before sending the frame, the source station senses the medium by checking the energy level at the carrier frequency.

a) Channel uses persistent strategy with back off until channel is idle.

b) After the station is found idle, the station waits for a period of time called DCF inter frame space(DIFS), then frame sends a control frame called request to send(RTS).

2. After receiving the RTS & waiting a period time called short inter frame space(SIFS), the destination station sends the a control frame, called clear to send(CTS)

Frame exchange Time Line

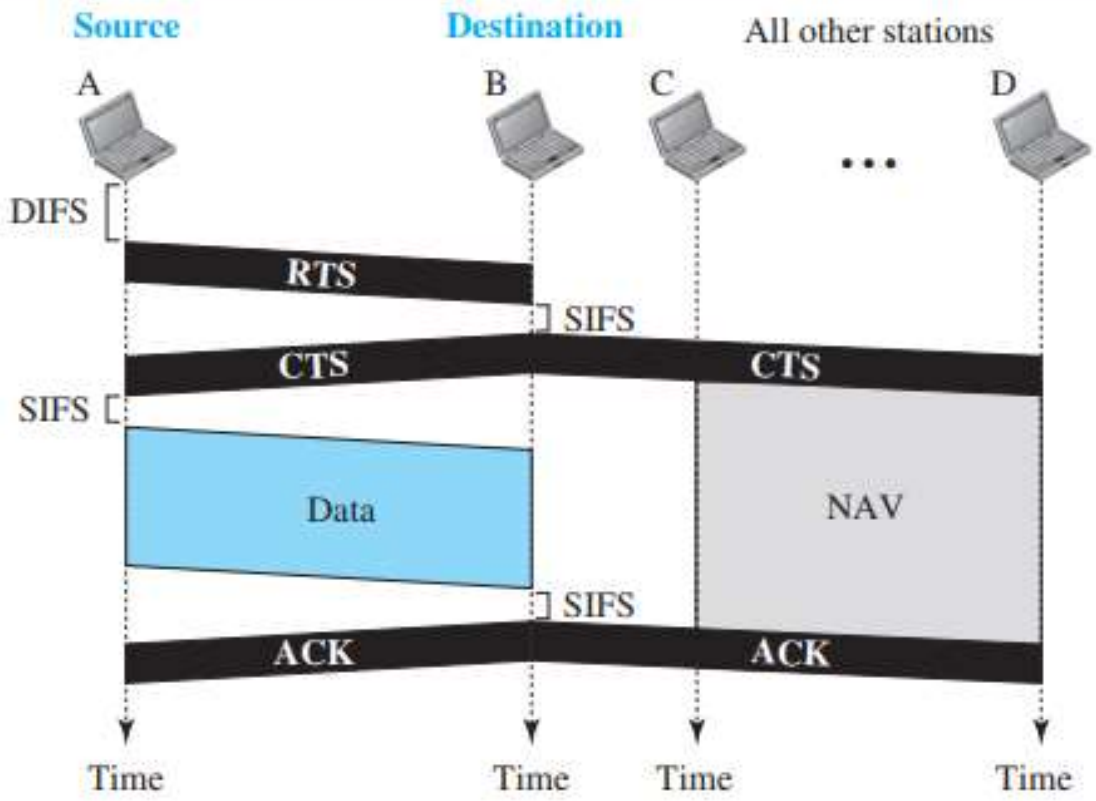
to the source station. The control frame indicates that the destination is ready to receive data.

3. Source station sends data after waiting an amount of time equal to SIFS.
4. Destination station after waiting an amount of time called SIFS, sends an acknowledgment to show that frame has been received.

Acknowledgment is needed in this protocol to check for the successful transmission of its data at the destination.

CSMA/CA and NAV

Figure 12.17 *CSMA/CA and NAV*



Network Allocation vector

- How do other stations defer sending data when one of the station acquire access i.e How collision avoidance aspect is accomplished??
- Feature is NAV

I. When station sends an RTS frame, it includes the duration of the time that it needs to occupy the channel.

- The stations that are affected by this transmission creates a times called Network Allocation Vector(NAV), that shows how much time is passed and when these stations are allowed to check the idleness of the medium.
- Each time a station access the system and sends an RTS frame, other stations starts their NAV.
- i.e Each station before sensing the medium to see if it is idle, it first checks NAV if it has expired.

Collision during the Handshaking

I. What happens if there is a collision during the time when RTS or CTS control frames are in transition(Handshaking period)

- Two or more stations try to sends RTS frame at the same time, the control frames may collide.
- Since there is no mechanism for collision detection, the sender assumes there has been a collision if CTS frame is not received from the receiver.
- The back off strategy is applied and sender tries again.

Hidden station Problem

- Solution is to use hand shake frames.
- When RTS message is sent from B to A, not C.
- Because both B & C are with in transmission range of A, the CTS message, which contains the duration of data transmission from B to A, reaches C.
- Station C knows that there is some hidden station is using the channel and refrains from transmitting until the duration is over.